Positive and Negative Transfer in Conditional Diffusion Models

Abstract

Joint conditional training and target-only training estimate the same score for a target class, yet they use different data and parameter-sharing structures. We study when auxiliary classes improve target-score estimation and when sharing is harmful. A Gaussian latent-factor model gives explicit risk formulas: compatible classes reduce covariance-estimation variance, whereas incompatible covariances introduce persistent target bias. For regular, fixed-dimensional denoising score-matching models, we separate shared-parameter error from class-specific error under equal-target and equal-total sample budgets. This distinction explains why an advantage under equal target size need not survive a fixed total budget. We connect small generator deformations to an approximate shared-score representation and derive a local threshold comparing mismatch bias with variance reduction. Two constructions establish that equal-total improvement is nevertheless possible when auxiliary classes are more informative about relevant shared directions. Gaussian simulations illustrate the predicted dependence on compatibility and sample allocation. An ImageNet case study further shows that gains in weighted score risk, denoising error, and generated-image metrics need not agree. The results identify conditions for transfer in score estimation rather than a universal advantage of conditional generation.

1 Introduction

Diffusion models generate data by reversing a sequence of Gaussian corruptions using estimated score functions [6, 14–17]. Class-conditional models incorporate labels into a shared denoising network. When the goal is to generate one target class, an alternative is to train an unconditional model on that class alone. Both procedures aim at the same target score, $\nabla_x \log p_t(x \mid c_*)$. Their statistical difference is how they estimate it.

Auxiliary classes can help estimate common features of the score field. Sharing can also impose an inaccurate common structure and bias the target predictor. Whether joint training helps therefore depends on the compatibility of the classes, the fitted score family, and the available sample budget. In particular, adding auxiliary observations while keeping the target sample fixed is a different comparison from replacing target observations with auxiliary observations under a fixed total budget.

We study these comparisons through a shared latent-generator model and regular parametric denoising score matching (DSM). A Gaussian example first makes the bias–variance tradeoff explicit. We then connect small generator deformations to an approximate shared-score representation and derive expected-risk expansions that separate shared and class-specific estimation error. Translation families show why pooling helps under equal target size but can hurt under equal total size. Two further examples demonstrate equal-total improvement when auxiliary observations provide more information about relevant shared directions. Finally, fixed and local incompatibility yield persistent bias and a quantitative transfer threshold, respectively.

Our theoretical conclusions concern target-score estimation in identifiable, fixed-dimensional models. They do not assert that arbitrary neural networks satisfy the required regularity conditions or that lower score risk orders all sampling metrics. The experiments examine these distinctions using Gaussian data and a real-image case study. The Gaussian simulations isolate

covariance compatibility and sample allocation; the image results compare weighted denoising error with KID and FID.

Notation and comparison. Let $p_0^c = p_0(\cdot \mid c)$ and p_t^c denote the original and noised class-c distributions. The pooled distribution is $p_t = \sum_c \pi_c p_t^c$, where $\pi_c = \mathbb{P}(C = c)$, and $s_c(x,t) = \nabla_x \log p_t^c(x)$. The target class is c_*, abbreviated by a subscript $*$ in class-indexed quantities. We distinguish population class weights π_c, training weights ρ_c, and limiting sample proportions κ_c. Positive transfer means strictly smaller expected target-score risk for joint training than for target-only training under the stated sample budget.

2 Model

We model class-c observations as

$$X_i^c = G_c(Z_i^c) + \tau \xi_i^c, \qquad G_c = G_0 + S_c, \qquad Z_i^c \overset{\text{iid}}{\sim} P_Z, \qquad \xi_i^c \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d). \tag{2.1}$$

All latent variables and noises are independent across observations and classes, and Z_i^c is independent of ξ_i^c. The latent dimension is $m \ll d$, the latent law P_Z is common across classes, and the observation-noise level $\tau \geq 0$ is treated as known. The map G_0 describes shared structure, while S_c describes a class-specific deformation. Shared latent structure has precedents in deformable templates, identity models, shared latent representations, and dictionary learning [1, 8, 9, 13, 19].

When G_c is an embedding of an m-dimensional latent domain, its image is an m-dimensional manifold. For $\tau = 0$, observations lie on this image; for $\tau > 0$, they have a full-dimensional Gaussian-smoothed distribution. A latent representation alone does not guarantee that its image is an embedded manifold.

2.1 Diffusion, training, and target risk

The variance-preserving forward marginal is

$$X_t^c = \sqrt{\alpha_t} X_0^c + \sqrt{1-\alpha_t}\,\varepsilon \overset{d}{=} \sqrt{\alpha_t} G_c(Z) + \sigma_t \varepsilon', \qquad \sigma_t^2 = \alpha_t \tau^2 + 1 - \alpha_t, \tag{2.2}$$

where $0 < \alpha_t < 1$ and $\varepsilon \sim \mathcal{N}(0, I_d)$ is independent of the original observation. Here $\varepsilon' = (\sqrt{\alpha_t}\tau\xi + \sqrt{1-\alpha_t}\varepsilon)/\sigma_t$ is standard Gaussian and independent of Z; it is not independent of ξ and ε. Thus

$$p_t^c(x) = \int \varphi_{\sigma_t^2 I_d}\{x - \sqrt{\alpha_t} G_c(z)\}\, dP_Z(z), \quad s_c(x,t) = -\frac{x}{\sigma_t^2} + \frac{\sqrt{\alpha_t}}{\sigma_t^2}\mathbb{E}[G_c(Z) \mid X_t^c = x]. \tag{2.3}$$

Here φ_Σ is the density of $\mathcal{N}(0, \Sigma)$. At $t = 0$, the ordinary ambient score may fail to exist when $\tau = 0$ and the data lie on a lower-dimensional manifold. Positive observation noise removes this singularity. We work on a fixed interval $[\delta, T]$, with $0 < \delta < T < \infty$, so the scores are well defined in either case.

For a nonnegative integrable weight w with positive integral, define the class score risk

$$\mathcal{R}_c(f) = \int_\delta^T w(t)\mathbb{E}_{X_t \sim p_t^c} \|f(X_t, t) - s_c(X_t, t)\|^2 \, dt. \tag{2.4}$$

Evaluation observations are independent of the training data. For a fitted predictor $\widehat{s}$, $\mathcal{R}_c(\widehat{s})$ holds the predictor fixed; an outer expectation $\mathbb{E}\mathcal{R}_c(\widehat{s})$ also averages over the training sample.

Table 1: Parameter sharing in representative class-conditional denoisers.

Component	Role	Sharing
Convolution or attention layers	Extract and combine image features	Shared
Time encoder and conditioning maps	Modulate intermediate features	Shared
Output layers	Produce score or noise predictions	Shared
Class embedding e_c	Specify the class condition	Class specific

A noise predictor $\widehat{\varepsilon}_\theta$ induces the score predictor $s_\theta(x,t,c) = -\widehat{\varepsilon}_\theta(x,t,c)/\sqrt{1-\alpha_t}$. Its population noise-prediction loss is

$$\sum_{c=1}^{K} \rho_c \mathbb{E}_{t,X_0^c,\varepsilon} \left\|\widehat{\varepsilon}_\theta(X_t^c,t,c) - \varepsilon\right\|^2 = \sum_{c=1}^{K} \rho_c \mathbb{E}_{t,X_0^c,\varepsilon} (1-\alpha_t) \left\| s_\theta(X_t^c,t,c) + \frac{\varepsilon}{\sqrt{1-\alpha_t}} \right\|^2, \tag{2.5}$$

where $\rho_c > 0$, $\sum_c \rho_c = 1$, and t is sampled independently of the data and noise. The conditional-expectation identity

$$s_c(x,t) = \mathbb{E}\left[-\frac{\varepsilon}{\sqrt{1-\alpha_t}} \,\middle|\, X_t^c = x \right] \tag{2.6}$$

implies that the population minimizer is the true conditional score. Uniform sampling on $[\delta, T]$ makes unweighted noise MSE equivalent to score error weighted by $(1-\alpha_t)/(T-\delta)$, up to a parameter-independent term.

Target-only training uses only observations from c_*; joint training uses labeled observations from all classes. Each joint-training observation is evaluated at its actual label. Both procedures target s_{c_*}. Without parameter sharing or another coupling between classes, the joint objective separates into independent classwise problems and supplies no transfer mechanism.

2.2 Conditional networks and shared score structure

Representative conditional U-Nets and diffusion transformers use a shared denoising network modulated by a class embedding [4, 10–12]. Write

$$s_{\theta,E}(x,t,c) = F_\theta(x,t,e_c), \qquad E = (e_1,\dots,e_K)^\top \in \mathbb{R}^{K\times q}. \tag{2.7}$$

The shared map $F_\theta : \mathbb{R}^d \times [\delta,T] \times \mathbb{R}^q \to \mathbb{R}^d$ may include the conversion from noise or denoised-data prediction to a score. In this abstraction, θ contains the common image-processing, conditioning, and output parameters; only the rows of E are explicitly class specific. A network with P shared parameters therefore has $P + Kq$ parameters, rather than K separate sets of network parameters.

The embedding acts throughout the network. For example, a shared conditioning map can combine e_c with a time embedding and produce block-specific scale and shift vectors:

$$z(t,c) = \tau_\theta(t) + e_c, \qquad \widetilde{v}_\ell = \{1 + \gamma_\ell(z(t,c))\} \odot \mathrm{Norm}(v_\ell) + \beta_\ell(z(t,c)).$$

Adaptive group normalization in conditional U-Nets and adaptive layer normalization in diffusion transformers implement versions of this mechanism. The scale and shift depend on the class, but the maps producing them are shared.

A local expansion in the embedding motivates a shared-basis model. Fix e_0, set $u_c = e_c - e_0$, and suppose F_θ is differentiable in its last argument. Then

$$F_\theta(x,t,e_c) = h_\theta(x,t) + H_\theta(x,t)u_c + R_{\theta,c}(x,t), \quad h_\theta = F_\theta(\cdot,\cdot,e_0), \quad H_\theta = D_e F_\theta(\cdot,\cdot,e_0). \tag{2.8}$$

If $D_e F_\theta$ is $L_\theta(x,t)$-Lipschitz along the relevant embedding segment, then $\|R_{\theta,c}(x,t)\| \le L_\theta(x,t)\|u_c\|^2/2$. The adjustment is linear in class coordinates but can depend nonlinearly on the noisy input and time.

Motivated by this expansion, we study true scores of the form

$$s_c(x,t) = h_0(x,t) + H_0(x,t)u_c + r_c(x,t), \qquad H_0(x,t) \in \mathbb{R}^{d\times q}. \tag{2.9}$$

The columns of H_0 describe shared directions of class variation; r_c records departures from this representation. An approximate model controls

$$\sum_c \rho_c \int_\delta^T \frac{1-\alpha_t}{T-\delta} \, \|r_c(\cdot,t)\|^2_{L^2(p_t^c)} \, dt \le \epsilon^2_{\text{app}}. \tag{2.10}$$

This is a structural restriction on scores arising from a common diffusion, not a claim that arbitrary fields define valid densities. Its content is a low-dimensional class representation with controlled shared-function complexity.

We use the centering convention

$$\sum_c \pi_c u_c = 0, \qquad \sum_c \pi_c r_c = 0, \qquad h_0 = \sum_c \pi_c s_c. \tag{2.11}$$

It removes an additive ambiguity but leaves changes of basis in H_0 and u_c. This convention describes the true representation; it does not supply known class centers or impose population constraints on fitted parameters. Moreover, h_0 is generally different from the pooled score, which uses posterior class weights:

$$\nabla \log p_t(x) = \sum_c \mathbb{P}(C=c \mid X_t = x) s_c(x,t).$$

Auxiliary data can improve the estimation of h_0 and H_0, while target observations identify the target combination. Large residuals or insufficient class-specific flexibility can instead create bias.

2.3 A Gaussian latent-factor example

Consider the special case

$$X_i^c = \mu_c + B_c Z_i^c + \tau \xi_i^c, \quad Z_i^c \sim \mathcal{N}(0, I_m), \quad \Sigma_c = B_c B_c^\top + \tau^2 I_d, \tag{2.12}$$

where $\tau > 0$, $1 \le m < d$, and $\operatorname{rank}(B_c) = m$. The Gaussian latent law is specific to this example; it is separate from the compact-domain assumptions used later for the geometric expansion. We fit unrestricted class means and a shared covariance of the form $BB^\top + \tau^2 I_d$, allowing $\operatorname{rank}(B) \le m$. Define

$$V_{c,t} = \alpha_t \Sigma_c + (1-\alpha_t) I_d, \qquad s_c(x,t) = -V_{c,t}^{-1}(x - \sqrt{\alpha_t}\mu_c).$$

Let n_c independent observations come from each class, $n = \sum_c n_c$, and $\rho_c > 0$ with $\sum_c \rho_c = 1$. Write

$$\overline{X}_c = \frac{1}{n_c}\sum_i X_i^c, \qquad \widehat{C}_c = \frac{1}{n_c}\sum_i (X_i^c - \overline{X}_c)(X_i^c - \overline{X}_c)^\top, \qquad \widehat{C}_\rho = \sum_c \rho_c \widehat{C}_c.$$

For a positive semidefinite matrix C with descending eigenvalues $\lambda_j(C)$ and eigenvectors $v_j(C)$, set

$$\mathcal{T}_m(C) = \tau^2 I_d + \sum_{j=1}^m \{\lambda_j(C) - \tau^2\}_+ v_j(C) v_j(C)^\top. \tag{2.13}$$

Empirical DSM, with exact averaging over corruption noise and time, gives

$$\widehat{\mu}_{c,\mathrm{J}} = \overline{X}_c, \quad \widehat{\Sigma}_{\mathrm{J}} = \mathcal{T}_m(\widehat{C}_\rho), \qquad \widehat{\mu}_{*,\mathrm{T}} = \overline{X}_{*,M}, \quad \widehat{\Sigma}_{\mathrm{T}} = \mathcal{T}_m(\widehat{C}_{*,M}), \tag{2.14}$$

where M is the target-only sample size. The fitted target score is

$$\widehat{s}_E(x,t) = -\{\alpha_t \widehat{\Sigma}_E + (1-\alpha_t) I_d\}^{-1}(x - \sqrt{\alpha_t}\widehat{\mu}_{*,E}), \qquad E \in \{\mathrm{J},\mathrm{T}\}. \tag{2.15}$$

Thus auxiliary observations affect the covariance estimate, whereas the target mean uses target observations alone. Appendix A derives these estimators directly from DSM.

Compatible covariances. Suppose $\Sigma_c = \Sigma_*$ for every class. Keep d, m, K, τ fixed, let every n_c/n_* have a positive finite limit, and take fixed training weights or $\rho_c = n_c/n$. For continuous $0 < \alpha_t < 1$ on $[\delta, T]$, there are explicit constants $C_\mu, C_B > 0$ such that

$$\mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{J}}) = \frac{C_\mu}{n_*} + C_B \sum_c \frac{\rho_c^2}{n_c} + o(n^{-1}), \tag{2.16}$$

$$\mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{T}}) = \frac{C_\mu + C_B}{M} + o(n^{-1}), \qquad M \asymp n. \tag{2.17}$$

Their formulas appear in (A.5). For proportional weights and at least two classes, equal target size ($M = n_*$) gives

$$\mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{J}}) - \mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{T}}) = -C_B \left(\frac{1}{n_*} - \frac{1}{n}\right) + o(n^{-1}) < 0 \tag{2.18}$$

for sufficiently large samples. Under equal total size ($M = n$), the leading difference is instead $C_\mu(1/n_* - 1/n) > 0$: pooling cannot improve covariance estimation beyond n compatible target observations, but it leaves fewer observations for estimating the target mean.

Fixed covariance mismatch. If the true covariances differ, the population joint optimizer is

$$\Sigma_{\mathrm{J}} = \mathcal{T}_m\left(\sum_c \rho_c \Sigma_c\right), \qquad s_{\mathrm{J}}(x,t) = -\{\alpha_t \Sigma_{\mathrm{J}} + (1-\alpha_t) I_d\}^{-1}(x - \sqrt{\alpha_t}\mu_*). \tag{2.19}$$

Assume the pooled covariance has a strict eigengap between its mth and $(m+1)$st eigenvalues, with the mth greater than τ^2. Its target bias is

$$\mathcal{R}_*(s_{\mathrm{J}}) = \int_\delta^T w(t)\alpha_t^2 \left\| \{\alpha_t \Sigma_{\mathrm{J}} + (1-\alpha_t) I_d\}^{-1} (\Sigma_{\mathrm{J}} - \Sigma_*) V_{*,t}^{-1/2} \right\|_F^2 dt. \tag{2.20}$$

For fixed covariances and weights, the expected joint risk converges to this quantity and the expected target-only risk converges to zero. The bias is positive exactly when $\Sigma_{\mathrm{J}} \neq \Sigma_*$. Hence a fixed incompatible pooled covariance eventually produces negative transfer.

Local transfer threshold. Keep Σ_* fixed and allow $\max_c \|\Sigma_c - \Sigma_*\|_F = O(n_*^{-1/2})$. Under equal target size,

$$\mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{J}}) - \mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{T}}) = \mathcal{R}_*(s_{\mathrm{J}}) + C_B \left(\sum_c \frac{\rho_c^2}{n_c} - \frac{1}{n_*}\right) + o(n^{-1}), \tag{2.21}$$

with C_B evaluated at Σ_*. For proportional weights, joint training helps when the mismatch bias is smaller than $C_B(1/n_* - 1/n)$ by a fixed positive multiple of n^{-1}, and hurts when it is larger by such a margin. An equality at leading order does not determine the transfer sign.

3 Theoretical results

We use the general Gaussian corruption model

$$X_t^c = a_t X_0^c + b_t \varepsilon, \qquad \sigma_t^2 = a_t^2 \tau^2 + b_t^2, \qquad t \in [\delta, T]. \tag{3.1}$$

The schedules are continuous, $0 < a_t \leq 1$, and $b_t > 0$. On this compact interval, the corruption standard deviation is bounded above and away from zero. The variance-preserving model has $a_t = \sqrt{\alpha_t}$ and $b_t = \sqrt{1-\alpha_t}$; additive-noise coordinates have $a_t = 1$ and $b_t = t$, where t denotes the noise standard deviation.

For a generator G, let s_G be the exact score induced by $G(Z) + \tau\xi$ under (3.1), with the latent and noise laws fixed. A fitted family $s_{\theta,u}$ may parameterize score fields directly or arise from candidate generators $G_{\theta,u}$. Directly parameterized fields need not be scores of normalized densities at every parameter value. Correct specification means $s_{\theta_0,u_c^0} = s_c$ for all classes, a condition distinct from geometric approximation.

For a nonnegative integrable training weight ω with positive integral, define the loss for one original observation x_0 by

$$\ell(\theta, u; x_0) = \frac{1}{2}\int_\delta^T \omega(t)\mathbb{E}_\varepsilon \left\| s_{\theta,u}(a_t x_0 + b_t\varepsilon, t) + \frac{\varepsilon}{b_t}\right\|^2 dt. \tag{3.2}$$

Joint and target-only fits minimize

$$\widehat{\mathcal{L}}_{\mathrm{J}} = \sum_c \frac{\rho_c}{n_c}\sum_{i=1}^{n_c} \ell(\theta, u_c; X_i^c), \qquad \widehat{\mathcal{L}}_{\mathrm{T}} = \frac{1}{M}\sum_{i=1}^{M} \ell(\theta, u; X_i^*), \tag{3.3}$$

and have the same attainable target-score family. Write $\widehat{s}_{\mathrm{J}}$ and $\widehat{s}_{\mathrm{T}}$ for their target predictors, and $\mathcal{R}_c(\theta, u) = \mathcal{R}_c(s_{\theta,u})$. Training and evaluation weights, ω and w, may differ. Uniform-time noise prediction in (2.5) uses $\omega(t) = (1-\alpha_t)/(T-\delta)$.

3.1 From shared generators to shared scores

Assumption 1 (Latent geometry)**.** The latent domain $\mathcal{Z} \subset \mathbb{R}^m$ is compact, P_Z is common to all classes, and G_0 extends to a C^2 map on a neighborhood of $\mathcal{Z}$. For constants $0 < L < U < \infty$,

$$L \le \lambda_{\min}\{J_{G_0}(z)^\top J_{G_0}(z)\} \le \lambda_{\max}\{J_{G_0}(z)^\top J_{G_0}(z)\} \le U, \qquad z \in \mathcal{Z}.$$

These bounds provide local nondegeneracy, not global injectivity. When a manifold interpretation is required, the relevant generators are additionally assumed to be embeddings. The score expansion below uses their boundedness and smoothness.

Assumption 2 (Small shared deformations)**.** For a C^2 field $D : \mathcal{Z} \to \mathbb{R}^{d\times q}$,

$$S_c(z) = D(z)u_c + R_c(z), \quad \|D\|_{C^2} \le L_D, \quad \max_c \|u_c\| \le \epsilon, \quad \max_c \|R_c\|_{C^2} \le \eta, \quad \sum_c \pi_c u_c = 0.$$

The coordinate normalization and bounds are fixed independently of sample size. The C^2 norm controls the map and its first two derivatives. The generator remainder R_c is distinct from the score residual r_c.

Thus $\max_c \|G_c - G_0\|_{C^2} \le L_D\epsilon + \eta$, controlling displacements and tangent and curvature changes.

For a bounded deformation V, differentiation of the Gaussian-smoothed score gives

$$\begin{aligned}\dot{s}_G[V](x,t) &:= \frac{d}{dv} s_{G+vV}(x,t)\Big|_{v=0} \\ &= \frac{a_t}{\sigma_t^2}\mathbb{E}[V(Z) \mid X_t = x] + \frac{a_t^2}{\sigma_t^4}\operatorname{Cov}\Big(G(Z), \{x - a_t G(Z)\}^\top V(Z) \,\Big|\, X_t = x\Big).\end{aligned} \tag{3.4}$$

The covariance term records the change in the latent posterior, which cannot generally be ignored.

Proposition 1 (Shared-score approximation)**.** *Under Assumptions 1–2, take $H_{0,j} = \dot{s}_{G_0}[D_j]$, where D_j is the jth column of D, and $h_0 = \sum_c \pi_c s_c$. Then (2.9) holds with*

$$\sum_c \pi_c r_c = 0, \qquad \max_c \int_\delta^T w(t)\,\|r_c(\cdot,t)\|^2_{L^2(p_t^c)}\,dt \le C_{\delta,w}(\epsilon^4 + \eta^2). \tag{3.5}$$

The same conclusion holds with ω in place of w. The constants are uniform over bounded families with fixed generator, deformation, schedule, and weight bounds.

The proposition supplies an approximation bound in the appropriate score norm. It does not establish that a fitted family represents the true scores exactly, nor does it by itself imply positive transfer.

3.2 DSM estimation and compatible sharing

We compare two sample budgets. Under *equal target size*, $M = n_*$: auxiliary observations are added without reducing the target sample. Under *equal total size*, $M = n$: auxiliary observations replace target observations.

Assumption 3 (Regular fixed-dimensional fitting)**.** Original observations are independent within and across classes. The joint and target-only problems admit identifiable, fixed-dimensional parameter charts; K is fixed. As $n \to \infty$, $n_c/n \to \kappa_c > 0$, $M \asymp n$, and the deterministic training weights have positive limits. Population minimizers are unique, interior, and separated, with positive definite Hessians. The compactness, smoothness, uniform moment, and expected-risk conditions in Appendix B.2 hold, uniformly for local alternatives.

Identifiability concerns statistical coordinates rather than raw neural-network weights. The exact noise and time averages in (3.2) also distinguish this analysis from an analysis of stochastic optimization.

The leading coefficients have a useful profiled form. At the correctly specified parameters, partition each class Hessian and one-observation loss gradient as

$$H_c = \begin{pmatrix} A_c & F_c \\ F_c^\top & C_c \end{pmatrix}, \qquad \psi_c = \begin{pmatrix} \psi_{\theta,c} \\ \psi_{u,c} \end{pmatrix}.$$

Eliminating the class coordinates gives

$$D_c = A_c - F_c C_c^{-1} F_c^\top, \qquad \zeta_c = \psi_{\theta,c} - F_c C_c^{-1} \psi_{u,c}, \qquad K_c = \mathrm{Var}(\zeta_c). \tag{3.6}$$

Let $D = \sum_c \kappa_c D_c$, $K = \sum_c \kappa_c K_c$, and $\Omega_{uu,*} = \mathrm{Var}(\psi_{u,*})$. When the relevant curvatures are positive definite, define

$$\mathcal{C}_{\mathrm{T}} = \mathrm{tr}(D_*^{-1} K_*), \qquad \mathcal{C}_{\mathrm{J}} = \mathrm{tr}(D_* D^{-1} K D^{-1}), \qquad \mathcal{C}_{\mathrm{cl}} = \mathrm{tr}(C_*^{-1} \Omega_{uu,*}). \tag{3.7}$$

These nonnegative quantities measure shared-parameter error for the two fits and the target class-coordinate error. Their derivation does not assume an information identity between gradient covariance and Hessian.

Theorem 1 (Risk under both sample budgets)**.** *Suppose $s_c = s_{\theta_0, u_c^0}$ for every class and Assumption 3 holds. Let $w = \omega$, $\rho_c = n_c/n$, and suppose $C_c \succ 0$, $D_* \succ 0$, and $D \succ 0$. Then*

$$\begin{aligned} \mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{J}}) &= \frac{\mathcal{C}_{\mathrm{J}}}{n} + \frac{\mathcal{C}_{\mathrm{cl}}}{n_*} + o(n^{-1}), \\ \mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{T}}) &= \frac{\mathcal{C}_{\mathrm{T}} + \mathcal{C}_{\mathrm{cl}}}{M} + o(n^{-1}). \end{aligned} \tag{3.8}$$

Consequently,

$$\mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{J}}) - \mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{T}}) = \frac{\mathcal{C}_{\mathrm{J}}}{n} - \frac{\mathcal{C}_{\mathrm{T}}}{M} + \mathcal{C}_{\mathrm{cl}} \left(\frac{1}{n_*} - \frac{1}{M} \right) + o(n^{-1}). \tag{3.9}$$

The coefficient of n^{-1} is $\mathcal{C}_{\mathrm{J}} - \kappa_*^{-1} \mathcal{C}_{\mathrm{T}}$ under equal target size and $\mathcal{C}_{\mathrm{J}} - \mathcal{C}_{\mathrm{T}} + (\kappa_*^{-1} - 1)\mathcal{C}_{\mathrm{cl}}$ under equal total size. A strictly negative coefficient implies positive transfer for all sufficiently large n; a strictly positive coefficient implies negative transfer. A zero coefficient leaves the comparison unresolved.

Under equal total size, the reduction in shared-parameter error must exceed the cost of fewer target observations. Appendix B.5 gives directional sufficient conditions. Appendix B.3 gives the general risk comparison for unequal training and evaluation weights and nonproportional class weights.

Corollary 1 (Translation families)**.** *Consider exact scores induced by*

$$X_0^c = G_\theta(Z) + \mathsf{T}u_c + \tau\xi,$$

where $\mathsf{T} \in \mathbb{R}^{d\times q}$ *is fixed with full column rank and shape and location are locally identifiable. Under correct specification, Assumption 3, and* $w = \omega$*,*

$$\mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{J}}) = \mathcal{C}_{\mathrm{sh}} \sum_c \frac{\rho_c^2}{n_c} + \frac{\mathcal{C}_{\mathrm{cl}}}{n_*} + o(n^{-1}), \qquad \mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{T}}) = \frac{\mathcal{C}_{\mathrm{sh}} + \mathcal{C}_{\mathrm{cl}}}{M} + o(n^{-1}). \tag{3.10}$$

The constants are independent of class centers. The conclusion also holds for $w \neq \omega$ *if* $Z \overset{d}{=} -Z$ *and* $G_\theta(-z) = -G_\theta(z)$ *throughout the local family.*

For proportional weights, $K \geq 2$, and $\kappa_* < 1$, $\mathcal{C}_{\mathrm{sh}} > 0$ yields positive transfer under equal target size. Under equal total size, the leading gap is $\mathcal{C}_{\mathrm{cl}}(1/n_* - 1/n)$, which is positive if $\mathcal{C}_{\mathrm{cl}} > 0$. Translations alone therefore do not supply an equal-total advantage in this regular model.

3.3 Two mechanisms for equal-total improvement

Auxiliary classes can compensate for fewer target observations if they are more informative about shared directions that matter for the target. The next two constructions isolate distinct ways this can occur.

Proposition 2 (Geometric sensitivity)**.** *Let* $Z \sim \mathrm{Unif}[-1, 1]$*,* $\tau > 0$*, and*

$$G_{\theta,u}(z) = (z, u\theta_1 f_\lambda(z), \ldots, u\theta_4 f_\lambda(z), u), \qquad f_\lambda(z) = 1 + \lambda z^2,$$

with $\theta \in \mathbb{R}^4$ *shared and* u *fitted separately for each class. The latent law and* λ *are fixed. Take three classes with true coordinates* $(u_1^0, u_2^0, u_3^0) = (1, 2, 2)$ *and target class 1. Joint fitting uses* $n/3$ *observations per class with weights* $1/3$*; target-only fitting uses* n *target observations. Under Assumption 3, joint fitting has strictly smaller expected target risk for all sufficiently large* n *in a neighborhood of* $(\lambda, \theta_0) = (0, 0)$*. At this reference point,*

$$\mathbb{E}\mathcal{R}_1(\widehat{s}_{\mathrm{J}}) = \frac{13C}{3n} + o(n^{-1}), \quad \mathbb{E}\mathcal{R}_1(\widehat{s}_{\mathrm{T}}) = \frac{5C}{n} + o(n^{-1}), \quad C = \tau^2 \int_\delta^T w(t)\frac{a_t^2}{\sigma_t^4}\, dt > 0. \tag{3.11}$$

The neighborhood includes curved embedded intervals with $\lambda > 0$ *and* $\theta_0 \neq 0$*. Scaling the true coordinates to* $(r, 2r, 2r)$ *for sufficiently small fixed* $r > 0$ *meets any prescribed small-deformation radius while preserving the reference-point risk constants. The neighborhood may depend on* r*. Matching weights* $w = \omega$ *are not required.*

The auxiliary classes amplify the shared shape directions. Their class coordinates are estimated, and the latent variables are unobserved; the gain does not rely on supplying either to the estimator.

For a second construction, let $X_0^c \sim \mathcal{N}(\mu_c, \lambda_0)$, with $\lambda_0 > 0$ and $\mu_* = 0$. Set $v_t(\lambda) = a_t^2\lambda + b_t^2$ and fit

$$s_{\theta,u_c}(x, t) = -\frac{x - a_t u_c}{v_t(\lambda)} + \beta f(t)\{x^2 - a_t^2 u_c^2 - v_t(\lambda)\}, \qquad \theta = (\lambda, \beta)^\top. \tag{3.12}$$

Here f is fixed and continuous. The truth is $\theta_0 = (\lambda_0, 0)$ and $u_c^0 = \mu_c$. Both fits estimate β, even though its true value is zero. For $\beta \neq 0$, the quadratic field need not be the score of a normalized density; it is an admissible DSM predictor.

Theorem 2 (A shared quadratic score direction)**.** *Fit* (3.12) *by* (3.3)*, with proportional weights and compact parameter intervals containing the truth in their interiors, with λ bounded away from zero. Suppose $\int \omega(t) f(t)^2\, dt > 0$, $n_c/n \to \kappa_c > 0$, and $\sum_c \kappa_c \mu_c = 0$. Under equal total size,*

$$\begin{aligned} \mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{T}}) &= \frac{C_\lambda + C_{\beta,\mathrm{T}} + \mathcal{C}_{\mathrm{cl}}}{n} + o(n^{-1}), \\ \mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{J}}) &= \frac{C_\lambda + C_{\beta,\mathrm{J}} + \kappa_*^{-1}\mathcal{C}_{\mathrm{cl}}}{n} + o(n^{-1}), \end{aligned} \tag{3.13}$$

with coefficients in (B.20)*. Hence positive transfer holds if*

$$C_{\beta,\mathrm{T}} - C_{\beta,\mathrm{J}} > (\kappa_*^{-1} - 1)\mathcal{C}_{\mathrm{cl}}. \tag{3.14}$$

If there are at least two auxiliary classes and $C_{\beta,\mathrm{T}} > (\kappa_*^{-1} - 1)\mathcal{C}_{\mathrm{cl}}$, this inequality is attained by a sufficiently large, finite weighted spread $\sum_c \kappa_c \mu_c^2$, while retaining zero weighted mean. The means and parameter intervals are fixed before $n \to \infty$.

The variance-parameter contribution cancels. The improvement comes from estimating an additional shared score direction more accurately, enough to offset the target-mean penalty. It is absent from the Gaussian mean-and-covariance-only comparison.

3.4 Incompatibility and the transfer threshold

Suppose the target score is representable as $s_* = s_{\theta_0, u_*^0}$, but auxiliary scores need not be representable at the same θ_0. Let s_{J} be the target predictor at the limiting joint population DSM minimizer.

Proposition 1 bounds the approximation component of this bias. If the fitted family jointly represents $h_0 + H_0 u_c$ for all classes and $w \le L_w \omega$ for some finite L_w, then

$$\mathcal{R}_*(s_{\mathrm{J}}) \le \frac{L_w C_{\delta,\omega}}{\overline{\rho}_*}(\epsilon^4 + \eta^2), \qquad \overline{\rho}_* := \lim_{n\to\infty} \rho_* > 0. \tag{3.15}$$

This bound requires joint representability of the approximating fields, not merely the existence of a shared-score expansion.

Theorem 3 (Fixed and local incompatibility)**.** *Assume target correct specification and Assumption 3.*

Fixed mismatch. *For fixed class distributions,*

$$\mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{J}}) \longrightarrow \mathcal{R}_*(s_{\mathrm{J}}), \qquad \mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{T}}) \longrightarrow 0. \tag{3.16}$$

If $\mathcal{R}_*(s_{\mathrm{J}}) > 0$, joint fitting eventually has larger target risk. A sufficient condition is

$$\nabla_\theta \sum_c \overline{\rho}_c \inf_u \mathcal{L}_c(\theta, u)\Big|_{\theta=\theta_0} \neq 0, \tag{3.17}$$

provided the classwise minimizers in u are unique, interior, and smooth near θ_0, and zero target risk requires $\theta = \theta_0$. Here $\mathcal{L}_c = \mathbb{E}\ell(\theta, u; X_0^c)$ and $\overline{\rho}_c = \lim \rho_c$.

Local mismatch. *Start from a correctly shared baseline $s_c = s_{G_c} = s_{\theta_0, u_c^0}$ with bounded C^2 generators on compact $\mathcal{Z}$. Perturb only the auxiliary generators:*

$$G_{c,n} = G_c + h_n V_c, \qquad V_* = 0, \qquad h_n = O(n^{-1/2}), \tag{3.18}$$

where the V_c are fixed bounded C^2 fields and the latent and noise laws remain fixed. For manifold models, the perturbed generators remain embeddings. Assume the statistical conditions hold uniformly. Let $s_{\mathrm{J},n}$ be the target predictor at the perturbed joint population minimizer. Then

$$\mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{J}}) - \mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{T}}) = \mathcal{R}_*(s_{\mathrm{J},n}) + V_{\mathrm{J},n} - V_{\mathrm{T},M} + o(n^{-1}), \tag{3.19}$$

where $\mathcal{R}_*(s_{\mathrm{J},n}) = O(h_n^2)$ and the variance terms are evaluated at the baseline, as defined in (B.12)*. For a fixed $\gamma > 0$, positive transfer follows if $\mathcal{R}_*(s_{\mathrm{J},n}) + V_{\mathrm{J},n} \le V_{\mathrm{T},M} - \gamma/n$ eventually; the reversed inequality with $+\gamma/n$ gives negative transfer.*

The local result compares bias and variance on the same n^{-1} scale. Appendix B.10 expresses the bias through the score derivative (3.4). In the translation family, suppose class c uses shape $\theta_0 + h_n a_c$, with $a_* = 0$, while the fitted family retains one shared shape. For $w = \omega$,

$$\mathcal{R}_*(s_{\mathrm{J},n}) = h_n^2 \left(\sum_c \rho_c a_c \right)^\top D \left(\sum_c \rho_c a_c \right) + o(h_n^2), \tag{3.20}$$

where D is the common profiled shape curvature. Optimizing the class coordinates is already accounted for in D.

Noise-level weights are another part of the comparison. Appendix B.11 constructs a misspecified family in which joint fitting decreases target score risk under one weight while increasing the target DSM loss under another. Noise prediction, denoising, and score estimation share a DSM formulation, but generally correspond to different weights. The conversions and extensions to DDPM and EDM noise domains appear in Appendix B.12. All proofs are given in Appendix B.

4 Numerical experiments

We compare joint conditional training (C_{J}) with target-only conditional training (C_{T}) under the two sample budgets in Section 3.2. Gaussian experiments also include a target-only unconditional control (U_{T}). The procedure labels C_{J} and C_{T} are distinct from the theoretical coefficients $\mathcal{C}_{\mathrm{J}}$ and $\mathcal{C}_{\mathrm{T}}$. All metrics concern the target class, and update counts are matched within each experiment. Appendices C–F give the protocols and supplementary results.

4.1 Gaussian classes with rotated covariances

We use three zero-mean Gaussian classes in $d = 100$, each with 50 covariance eigenvalues equal to 1.8 and 50 equal to 0.2. The auxiliary eigenspaces are rotated by $+\theta$ and $-\theta$ relative to the target, with $\theta \in \{0^\circ, 30^\circ, 45^\circ, 75^\circ\}$. This varies orientation while preserving each spectrum. At zero rotation, all classes coincide. The model is a special case of (2.12), with $m = 50$ and $\tau^2 = 0.2$.

Joint training uses $n_* \in \{200, 600, 1800, 5400\}$ observations per class, so $n = 3n_*$. DDPM noise-prediction MLPs are trained for 20,000 updates over 20 paired seeds at standard and limited capacities. Within each capacity, C_{T} and C_{J} use the same architecture, including the full three-class embedding table. The label-free U_{T} provides a separate parameterization control.

Under equal target size, standard-capacity joint training reduces score risk at every angle for $n_* = 200$ and 600. At each nonzero angle, the ordering reverses at $n_* = 1800$ and 5400 (Table 2; Figure 1). For example, at 45°, joint and target-only conditional risks are 31.56 and 75.55 when $n_* = 600$, but 21.59 and 18.88 when $n_* = 1800$. The crossover is consistent with a diminishing variance advantage and a persistent cost of incompatible sharing.

Under equal total size, joint training has higher score risk than both target-only controls at every nonzero angle. At zero rotation, C_{J} improves on C_{T} at the three larger sample sizes, whereas U_{T} has the lowest risk throughout. Thus architecture matching and sample-budget matching answer separate questions; a gain over one target-only control need not hold against the other.

The evaluation criterion also matters. At limited capacity and $n_* = 600$, zero-rotation joint training improves all three metrics. At 30° and 45°, its lower score risk coexists with higher noise-prediction MSE (Table 3). These criteria weight noise levels differently, as shown in

Table 2: Target score risk in the standard-capacity rotation experiment at 20,000 updates. Entries are means ± standard errors over 20 paired seeds. The joint model uses n_* observations per class; each joint column is compared with the target-only controls in the same row.

n_*	U_{T}	C_{T}	$C_{\mathrm{J}}(0°)$	$C_{\mathrm{J}}(30°)$	$C_{\mathrm{J}}(45°)$	$C_{\mathrm{J}}(75°)$
Equal target size: $M = n_*$						
200	1215.64 ± 9.12	1526.66 ± 8.87	80.91 ± 0.61	105.13 ± 0.87	106.46 ± 1.13	147.61 ± 2.02
600	65.47 ± 0.45	75.55 ± 0.61	18.47 ± 0.13	30.25 ± 0.21	31.56 ± 0.33	39.04 ± 0.54
1800	16.36 ± 0.14	18.88 ± 0.10	12.69 ± 0.10	21.03 ± 0.15	21.59 ± 0.31	21.80 ± 0.41
5400	12.06 ± 0.14	13.42 ± 0.11	11.64 ± 0.09	18.83 ± 0.16	19.24 ± 0.25	19.01 ± 0.37
Equal total size: $M = 3n_*$						
200	65.47 ± 0.45	75.55 ± 0.61	80.91 ± 0.61	105.13 ± 0.87	106.46 ± 1.13	147.61 ± 2.02
600	16.36 ± 0.14	18.88 ± 0.10	18.47 ± 0.13	30.25 ± 0.21	31.56 ± 0.33	39.04 ± 0.54
1800	12.06 ± 0.14	13.42 ± 0.11	12.69 ± 0.10	21.03 ± 0.15	21.59 ± 0.31	21.80 ± 0.41
5400	11.09 ± 0.12	12.41 ± 0.10	11.64 ± 0.09	18.83 ± 0.16	19.24 ± 0.25	19.01 ± 0.37

Table 3: Limited-capacity rotation experiment under equal target size, $n_* = M = 600$, at 20,000 updates. Entries are means ± standard errors over 20 seeds. Noise MSE is per coordinate; Gaussian W_2^2 compares generated moments with the target Gaussian. Lower values are better.

Model	Score risk	Noise MSE	Gaussian W_2^2
U_{T}	56.25 ± 0.39	0.25833 ± 0.00010	4.25 ± 0.08
C_{T}	65.42 ± 0.59	0.25860 ± 0.00011	4.24 ± 0.06
$C_{\mathrm{J}}(0°)$	37.37 ± 0.32	0.25301 ± 0.00008	3.10 ± 0.05
$C_{\mathrm{J}}(30°)$	49.72 ± 0.24	0.26825 ± 0.00027	10.39 ± 0.18
$C_{\mathrm{J}}(45°)$	51.45 ± 0.42	0.27775 ± 0.00017	17.29 ± 0.14
$C_{\mathrm{J}}(75°)$	72.52 ± 0.80	0.26575 ± 0.00048	7.17 ± 0.28

Appendix D.1. Moment-matched Gaussian W_2^2 measures recovery of generated means and covariances, not all departures from the target law. The AR(1) experiment in Appendix E gives further examples of differing metric rankings.

4.2 Real images

We use 64×64 ImageNet images with French bulldog as the target class. Compact EDM–NCSN++ models [7] are trained from scratch for 80,000 updates. We evaluate the target weighted denoising MSE under the EDM noise distribution, together with KID and FID [2, 5]. Population differences in the denoising criterion equal differences in a corresponding weighted score risk, up to the pixel normalization. All image results use one training seed and are descriptive comparisons; they do not establish statistical significance across training runs.

Equal target size. Joint training combines $n_* \in \{250, 500\}$ target images with n_* images from each of three auxiliary classes. Close auxiliaries are related dog breeds; far auxiliaries are non-dog classes. Both groups lower test denoising MSE at both sample sizes (Table 4). Validation-selected checkpoints retain this denoising advantage (Table 10).

The generated-image metrics give a more qualified comparison. At $n_* = 250$, both joint models have higher KID and FID than C_{T}, despite their lower denoising MSE. At $n_* = 500$, close auxiliaries improve all three metrics: denoising MSE, KID, and FID decrease by 26.0%, 34.8%, and 21.5%, respectively. Far auxiliaries lower denoising MSE but worsen KID and FID. Close auxiliaries outperform far auxiliaries in both generated-image metrics at each sample size. Figure 6 shows the first 64 generated images from the fixed saved sequence for the close-auxiliary model with $n_* = 500$.

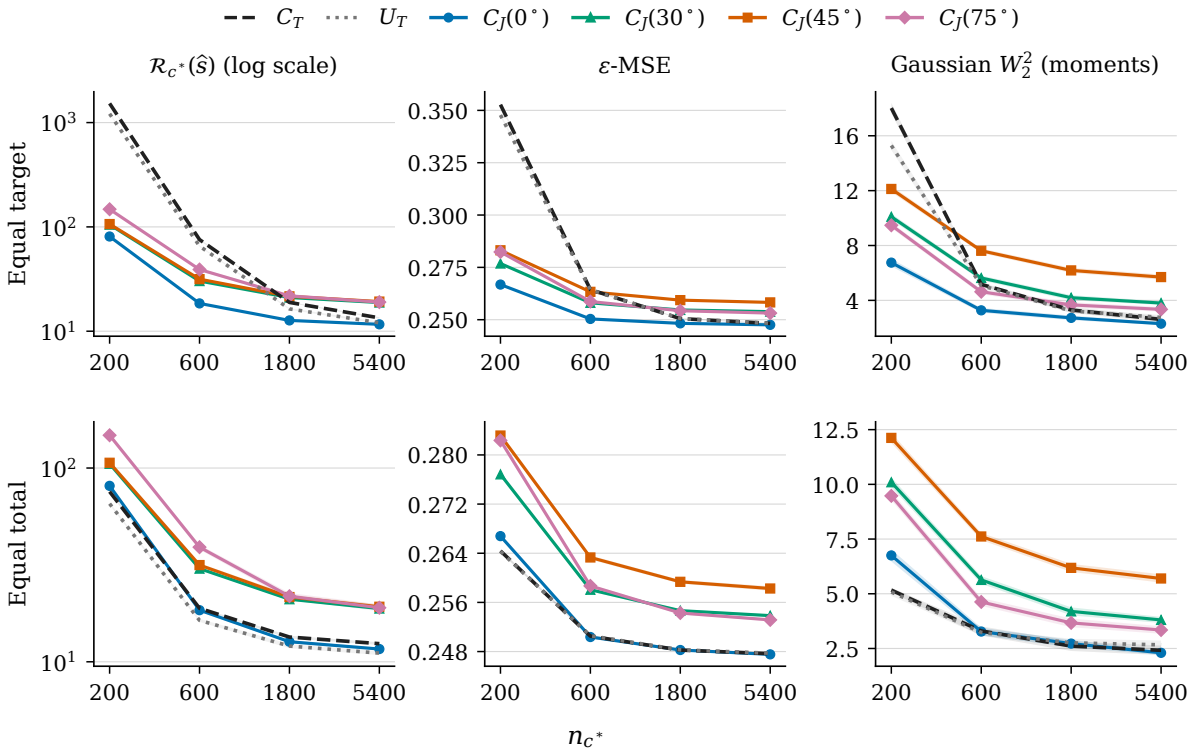


Figure 1: Standard-capacity rotation experiment. Rows compare equal target size (top) and equal total size (bottom); columns show target score risk, per-coordinate noise MSE, and moment-matched Gaussian W_2^2. Curves are means over 20 paired seeds; bands are pointwise 95% Student-t intervals. The horizontal axis is the joint target size n_*. Horizontal and score-risk axes are logarithmic. Lower values are better.

Table 4: ImageNet results under equal target size, using seed 0 and the 80,000-update EMA checkpoint. Evaluation uses 600 common target test images, eight paired noise draws per image, and 2,048 generated images. KID is multiplied by 10^3; FID is unscaled. Training size counts all classes.

Training size	Model	Denoising MSE	KID	FID
Target size $n_* = 250$				
250	C_{T}	0.231249	22.256	73.64
1000	$C_{\mathrm{J}}^{\mathrm{close}}$	0.150136	41.369	86.33
1000	$C_{\mathrm{J}}^{\mathrm{far}}$	0.149166	55.783	103.79
Target size $n_* = 500$				
500	C_{T}	0.185659	38.127	88.46
2000	$C_{\mathrm{J}}^{\mathrm{close}}$	0.137460	24.874	69.45
2000	$C_{\mathrm{J}}^{\mathrm{far}}$	0.137628	62.691	108.88

Table 5: ImageNet results under equal total size, $n = M = 1000$, using seed 0 and the 80,000-update EMA checkpoint. Evaluation uses 100 common target test images, two paired noise draws per image, and 2,048 generated images. KID is multiplied by 10^3; FID is unscaled.

Model	Target fraction	Denoising MSE	KID	FID
C_{T}	1.00	0.145891	26.433	130.98
$C_{\mathrm{J}}^{\mathrm{close}}$	0.85	0.145737	32.035	136.36
$C_{\mathrm{J}}^{\mathrm{close}}$	0.75	0.145286	37.102	140.05
$C_{\mathrm{J}}^{\mathrm{close}}$	0.50	0.146528	33.947	139.12

Equal total size. All models use 1,000 training images. Joint models allocate 85%, 75%, or 50% to the target, with the remainder split across the three close auxiliary classes. The 75% allocation has the lowest test denoising MSE, but all joint values are within 0.5% of C_{T} (Table 5). KID and FID favor the all-target allocation. These small denoising differences should not be read as evidence of a reproducible equal-total advantage from a single training seed.

Scope and interpretation. The theory identifies a target-score bias–variance tradeoff and the role of sample allocation. The experiments show that these factors also matter in neural training, while optimization, finite capacity, and metric weighting can affect the observed ordering. A smaller weighted score risk alone does not imply uniformly better KID, FID, or generated moments. Converting estimation comparisons into a sampling-error theorem would require additional control of initialization, reverse dynamics, and discretization. Extending the statistical results beyond regular fixed-dimensional charts, and replicating the image study across training seeds, remain separate tasks.

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A Proofs for the Gaussian example

Throughout the appendices, a class subscript $*$ abbreviates c_*. Samples are independent within and across classes. We keep d, m, K fixed, assume $\tau > 0$ is known and $1 \le m < d$, and let all class sample sizes grow in positive fixed proportions. Training averages exactly over corruption noise and time, and evaluation data are independent of training data.

A.1 The empirical DSM optimizer

For a candidate covariance $\Sigma = BB^\top + \tau^2 I_d$, with rank at most m above the noise floor, put $A_t = \{\alpha_t \Sigma + (1-\alpha_t) I_d\}^{-1}$. The class-$c$ score is $-A_t(x - \sqrt{\alpha_t}\mu_c)$. Integrating its noise-prediction loss over ε gives, for a fixed original observation,

$$\alpha_t(1-\alpha_t)(X_i^c - \mu_c)^\top A_t^2 (X_i^c - \mu_c) + \operatorname{tr}\{I_d - (1-\alpha_t)A_t\}^2.$$

Completing the square gives $\widehat{\mu}_c = \overline{X}_c$. The remaining pooled objective at time t is

$$(1-\alpha_t)\operatorname{tr}\Big[A_t^2\{\alpha_t \widehat{C}_\rho + (1-\alpha_t) I_d\}\Big] - 2(1-\alpha_t)\operatorname{tr}(A_t) + d. \tag{A.1}$$

The trace rearrangement inequality aligns the largest fitted covariance eigenvalues with the largest eigenvalues ℓ of $\widehat{C}_\rho$. In that basis, a fitted eigenvalue $\lambda \ge \tau^2$ contributes

$$f_t(\lambda; \ell) = \frac{(1-\alpha_t)(\alpha_t \ell + 1 - \alpha_t)}{(\alpha_t \lambda + 1 - \alpha_t)^2} - \frac{2(1-\alpha_t)}{\alpha_t \lambda + 1 - \alpha_t} + 1,$$

whose derivative is

$$\partial_\lambda f_t(\lambda; \ell) = \frac{2\alpha_t^2 (1-\alpha_t)(\lambda - \ell)}{(\alpha_t \lambda + 1 - \alpha_t)^3}.$$

An available direction is therefore fitted at $\max(\ell, \tau^2)$. For $\ell > \tau^2$, its improvement over the noise floor equals

$$\frac{(1-\alpha_t)\alpha_t^2 (\ell - \tau^2)^2}{(\alpha_t \tau^2 + 1 - \alpha_t)^2 (\alpha_t \ell + 1 - \alpha_t)},$$

which increases with ℓ. The rank constraint selects the leading m directions. These choices minimize the objective at every time, so time averaging preserves the optimizer. This proves (2.14); replacing empirical covariances by population covariances proves (2.19).

A.2 An exact risk decomposition

For either fitted covariance $\widehat{\Sigma}$, define

$$V_{*,t} = \alpha_t \Sigma_* + (1-\alpha_t) I_d, \quad A_{*,t} = V_{*,t}^{-1}, \quad \widehat{A}_t = \{\alpha_t \widehat{\Sigma} + (1-\alpha_t) I_d\}^{-1}.$$

Writing $Y_t = X_t^* - \sqrt{\alpha_t}\mu_*$, the score error is

$$\widehat{s}(X_t^*, t) - s_*(X_t^*, t) = -(\widehat{A}_t - A_{*,t}) Y_t + \sqrt{\alpha_t}\widehat{A}_t(\widehat{\mu}_* - \mu_*).$$

Conditional on the training data, Y_t is centered with covariance $V_{*,t}$, so the two terms have zero cross expectation. Gaussian sample means and centered sample covariances are independent, with

$$\overline{X}_c \sim \mathcal{N}(\mu_c, \Sigma_c / n_c), \qquad n_c \widehat{C}_c \sim W_d(\Sigma_c, n_c - 1)$$

[3]. Consequently,

$$\mathbb{E}\mathcal{R}_*(\widehat{s}) = \int_\delta^T w(t) \left[\mathbb{E}\operatorname{tr}\{(\widehat{A}_t - A_{*,t}) V_{*,t} (\widehat{A}_t - A_{*,t})\} + \frac{\alpha_t}{N_\mu}\operatorname{tr}\{\Sigma_* \mathbb{E}(\widehat{A}_t^2)\}\right] dt, \tag{A.2}$$

where $N_\mu = n_*$ for joint fitting and $N_\mu = M$ for target-only fitting. All fitted covariances satisfy $\widehat{\Sigma} \succeq \tau^2 I_d$, giving the uniform inverse bound $\left\|\widehat{A}_t\right\|_{\mathrm{op}} \le \min(\tau^2, 1)^{-1}$.

A.3 Compatible covariances

Let the common covariance be Σ, with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_m > \tau^2$ and $\lambda_{m+1} = \cdots = \lambda_d = \tau^2$. If Q projects onto the noise eigenspace, the derivative of (2.13) is

$$D\mathcal{T}_m(\Sigma)[H] = H - QHQ. \tag{A.3}$$

Indeed, near Σ, the positive leading singular values of $C - \tau^2 I_d$ are separated from the remaining singular values, so the truncated-SVD expansion applies [18]. For $H_\rho = \widehat{C}_\rho - \Sigma$,

$$\widehat{\Sigma}_{\mathrm{J}} - \Sigma = H_\rho - QH_\rho Q + R_n, \qquad \mathbb{E}\,\|R_n\|_F^2 = O(n^{-2}). \tag{A.4}$$

The local remainder is $O(\|H_\rho\|_F^2)$. Outside a fixed neighborhood, linear growth of $\mathcal{T}_m$ gives a bound of the same form after enlarging the constant. Fixed-dimensional Gaussian covariance moments then yield the displayed mean-square remainder and $\mathbb{E}\left\|\widehat{\Sigma}_{\mathrm{J}} - \Sigma\right\|_F^k = O(n^{-k/2})$ for $k = 2, 4$. The same bounds hold for the target-only fit.

In an eigenbasis of Σ, Wishart moments give

$$\mathbb{E}(H_\rho)_{ij}^2 = (\lambda_i\lambda_j + \mathbf{1}_{\{i=j\}}\lambda_i^2)\sum_c \frac{\rho_c^2}{n_c} + O(n^{-2}).$$

The resolvent identity, with $A_t = \{\alpha_t\Sigma + (1-\alpha_t)I_d\}^{-1}$ and $\Delta = \widehat{\Sigma} - \Sigma$, gives

$$\widehat{A}_t - A_t = -\alpha_t A_t \Delta A_t + O(\|\Delta\|_F^2)$$

uniformly in time. Cauchy–Schwarz and the preceding moment bounds make the omitted quadratic-risk terms $O(n^{-3/2})$. The leading covariance contribution is

$$\alpha_t^2 \mathbb{E}\,\mathrm{tr}(A_t\Delta A_t\Delta A_t), \qquad \mathrm{tr}(A_t\Delta A_t\Delta A_t) = \sum_{i,j} \frac{\Delta_{ij}^2}{(\alpha_t\lambda_i + 1 - \alpha_t)^2(\alpha_t\lambda_j + 1 - \alpha_t)}.$$

The derivative (A.3) retains exactly the entries with $i \leq m$ or $j \leq m$. Also, $\sup_t \left\|\mathbb{E}\widehat{A}_t^2 - A_t^2\right\|_F = O(n^{-1/2})$. Thus

$$\begin{aligned} C_\mu &= \int_\delta^T w(t)\alpha_t \sum_{i=1}^d \frac{\lambda_i}{(\alpha_t\lambda_i + 1 - \alpha_t)^2}\,dt, \\ C_B &= \int_\delta^T w(t)\alpha_t^2 \sum_{\substack{1\leq i,j\leq d \\ i\leq m \text{ or } j\leq m}} \frac{\lambda_i\lambda_j + \mathbf{1}_{\{i=j\}}\lambda_i^2}{(\alpha_t\lambda_i + 1 - \alpha_t)^2(\alpha_t\lambda_j + 1 - \alpha_t)}\,dt. \end{aligned} \tag{A.5}$$

Both constants are finite and positive. Substitution into (A.2) proves (2.16)–(2.17). The identity $\sum_c \rho_c^2/n_c = 1/n$ for proportional weights gives both budget comparisons in Section 2.3.

A.4 Fixed mismatch

Set $\Sigma_\rho = \sum_c \rho_c\Sigma_c$. The eigengap makes $\mathcal{T}_m$ continuous near Σ_ρ, so $\widehat{\Sigma}_{\mathrm{J}} \to \Sigma_{\mathrm{J}}$ and $\widehat{\Sigma}_{\mathrm{T}} \to \Sigma_*$ in probability. The uniform inverse bound in Appendix A.2 bounds the covariance-risk integrand deterministically, and the mean term is $O(N_\mu^{-1})$. Bounded convergence therefore gives the expected-risk limits.

Write $A_{\mathrm{J},t} = \{\alpha_t\Sigma_{\mathrm{J}} + (1-\alpha_t)I_d\}^{-1}$ and $\Delta = \Sigma_{\mathrm{J}} - \Sigma_*$. The resolvent identity yields

$$A_{\mathrm{J},t} - A_{*,t} = -\alpha_t A_{\mathrm{J},t}\Delta A_{*,t}.$$

Its quadratic risk equals (2.20). Since both matrix factors are invertible, $\alpha_t > 0$, and w has positive integral, the bias vanishes exactly when $\Delta = 0$. Moreover,

$$\mathcal{R}_*(s_{\mathrm{J}}) \leq \min(\tau^2, 1)^{-3}\,\|\Sigma_{\mathrm{J}} - \Sigma_*\|_F^2 \int_\delta^T w(t)\alpha_t^2\,dt. \tag{A.6}$$

A positive limiting bias gives eventual negative transfer.

A.5 Local mismatch

Keep $\Sigma = \Sigma_*$ fixed and suppose $\max_c \|\Sigma_c - \Sigma\|_F = O(n^{-1/2})$. Let $\Sigma_\rho = \sum_c \rho_c \Sigma_c$ and $U_n = \widehat{C}_\rho - \Sigma_\rho$, using the current covariances and weights. Smoothness of the spectral map gives

$$\|\Sigma_{\mathrm{J}} - \Sigma\|_F = O(n^{-1/2}), \qquad \mathcal{R}_*(s_{\mathrm{J}}) = O(n^{-1}),$$

and the localization argument above gives

$$\widehat{\Sigma}_{\mathrm{J}} - \Sigma_{\mathrm{J}} = D\mathcal{T}_m(\Sigma_\rho)[U_n] + R_n, \qquad \mathbb{E}\,\|R_n\|_F^2 = O(n^{-2}).$$

Gaussian covariance moments imply $\|\mathbb{E}U_n\|_F = O(n^{-1})$ and $\mathbb{E}\,\|U_n\|_F^k = O(n^{-k/2})$ for $k = 2, 4$. Further,

$$\begin{aligned}\mathrm{Cov}((U_n)_{ij}, (U_n)_{k\ell}) &= \sum_c \rho_c^2 \frac{n_c - 1}{n_c^2}\{(\Sigma_c)_{ik}(\Sigma_c)_{j\ell} + (\Sigma_c)_{i\ell}(\Sigma_c)_{jk}\} \\ &= \left(\sum_c \frac{\rho_c^2}{n_c}\right)(\Sigma_{ik}\Sigma_{j\ell} + \Sigma_{i\ell}\Sigma_{jk}) + O(n^{-3/2}).\end{aligned}$$

Because $D\mathcal{T}_m(\Sigma_\rho) = D\mathcal{T}_m(\Sigma) + O(n^{-1/2})$, the leading covariance-risk coefficient remains C_B at the target covariance.

Decompose $\widehat{A}_t - A_{*,t} = (A_{\mathrm{J},t} - A_{*,t}) + (\widehat{A}_t - A_{\mathrm{J},t})$. The first term is deterministic and $O(n^{-1/2})$; the second has mean $O(n^{-1})$ and second moment $O(n^{-1})$, uniformly in time. Their expected cross term is $O(n^{-3/2})$. The change in the mean-estimation coefficient is also negligible at order n^{-1}. Hence

$$\mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{J}}) = \mathcal{R}_*(s_{\mathrm{J}}) + \frac{C_\mu}{n_*} + C_B \sum_c \frac{\rho_c^2}{n_c} + o(n^{-1}).$$

Subtracting the unchanged target-only expansion with $M = n_*$ proves (2.21). A fixed positive margin on the n^{-1} scale determines the transfer sign.

B Proofs for the transfer results

B.1 Gaussian smoothing and shared-score approximation

We prove (3.4) and Proposition 1. Fix t, abbreviate $a = a_t$ and $\sigma = \sigma_t$, and consider

$$p_v(x) = \int \varphi_{\sigma^2 I_d}\{x - \mu_v(z)\}\, dP_Z(z), \qquad \mu_v = \mu_0 + v\beta, \qquad 0 \le v \le 1,$$

with $\sup_{v,z} \|\mu_v(z)\| \le M_0$ and $\sup_z \|\beta(z)\| \le B$. Let $\mathbb{E}_v$ and Cov_v refer to the posterior of Z given x under p_v, and set

$$A_v(z) = \frac{\{x - \mu_v(z)\}^\top \beta(z)}{\sigma^2}, \qquad A_v'(z) = -\frac{\|\beta(z)\|^2}{\sigma^2}.$$

Differentiating the posterior numerator and denominator gives

$$\frac{d}{dv}\mathbb{E}_v[f_v] = \mathbb{E}_v[f_v'] + \mathrm{Cov}_v(f_v, A_v). \tag{B.1}$$

The bounded centers and Gaussian kernel justify differentiation under the integral. The score is $s_v = (\mathbb{E}_v\mu_v - x)/\sigma^2$, so

$$\begin{aligned}s_v' &= \sigma^{-2}\{\mathbb{E}_v\beta + \mathrm{Cov}_v(\mu_v, A_v)\}, \\ s_v'' &= \sigma^{-2}\big[2\,\mathrm{Cov}_v(\beta, A_v) + \mathrm{Cov}_v(\mu_v, A_v') + \mathbb{E}_v\{(\mu_v - \mathbb{E}_v\mu_v)(A_v - \mathbb{E}_v A_v)^2\}\big].\end{aligned}$$

Taking $\mu_v = a(G + vV)$ proves (3.4). With $R_x = \|x\| + M_0$, direct bounds give

$$\|s_v'\| \le \frac{B}{\sigma^2}\left(1 + \frac{2M_0R_x}{\sigma^2}\right), \qquad \|s_v''\| \le B^2\left(\frac{6R_x}{\sigma^4} + \frac{8M_0R_x^2}{\sigma^6}\right). \tag{B.2}$$

Apply these inequalities to the paths G_0+vDu_c and $G_0+Du_c+vR_c$, with $R = \|G_0\|_\infty+L_D\epsilon+\eta$. Their Gaussian centers have norm at most a_tR, and the corresponding deformation bounds are $a_tL_D\epsilon$ and $a_t\eta$. By linearity, $\dot{s}_{G_0}[Du_c] = H_0u_c$. Taylor expansion along the first path and integration along the second give the uncentered remainder $\widetilde{r}_c = s_c - s_{G_0} - H_0u_c$ and the bound

$$\begin{aligned}\|\widetilde{r}_c(x,t)\| \le a_t^2L_D^2\epsilon^2\left\{\frac{3(\|x\| + a_tR)}{\sigma_t^4} + \frac{4a_tR(\|x\| + a_tR)^2}{\sigma_t^6}\right\}\\ + \frac{a_t\eta}{\sigma_t^2}\left\{1 + \frac{2a_tR(\|x\| + a_tR)}{\sigma_t^2}\right\}.\end{aligned} \tag{B.3}$$

The positive noise lower bound reduces this to a common envelope $C\{\epsilon^2(1+\|x\|^2)+\eta(1+\|x\|)\}$. Every class marginal has uniformly bounded fourth moments because its generator is bounded and its noise is Gaussian. Thus, for all j, k,

$$\int_\delta^T w(t)\,\|\widetilde{r}_j(\cdot,t)\|^2_{L^2(p_t^k)}\,dt \le C_{\delta,w}(\epsilon^4+\eta^2).$$

Set $\overline{r} = \sum_j \pi_j\widetilde{r}_j$ and $r_c = \widetilde{r}_c - \overline{r}$. Centering of u_c gives $h_0 = \sum_j \pi_j s_j = s_{G_0} + \overline{r}$ and $\sum_c \pi_c r_c = 0$. Jensen's inequality, $\|r_c\|^2 \le 2\|\widetilde{r}_c\|^2 + 2\sum_j \pi_j\|\widetilde{r}_j\|^2$, then proves (3.5). Replacing w by ω proves the training-weighted version.

B.2 DSM identities and statistical regularity

Population identities. For $X_t^c = a_tX_0^c + b_t\varepsilon$, differentiation of the Gaussian convolution gives

$$s_c(x,t) = \mathbb{E}[-\varepsilon/b_t \mid X_t^c = x]. \tag{B.4}$$

Orthogonality of conditional expectation yields

$$\mathcal{L}_c(\theta,u) := \mathbb{E}\ell(\theta,u;X_0^c) = \frac{1}{2}\int_\delta^T \omega(t)\,\|s_{\theta,u}(\cdot,t) - s_c(\cdot,t)\|^2_{L^2(p_t^c)}\,dt + \text{constant}, \tag{B.5}$$

where the constant does not depend on the fitted parameters. If $w = \omega$, population DSM and score risk therefore have the same minimizers.

Let $\nu(t)$ be the noise-level sampling density. A noise predictor $\widehat{\varepsilon}_{\theta,u} = -b_ts_{\theta,u}$ has DSM weight

$$\omega(t) = \nu(t)b_t^2. \tag{B.6}$$

For a denoiser $D_{\theta,u}$ predicting X_0^c, set

$$s_{\theta,u}(x,t) = \frac{a_tD_{\theta,u}(x,t) - x}{b_t^2}.$$

Then $s_{\theta,u}(X_t^c,t)+\varepsilon/b_t = (a_t/b_t^2)\{D_{\theta,u}(X_t^c,t)-X_0^c\}$. A denoising loss weighted by $\lambda_D(t)$ therefore has DSM weight

$$\omega(t) = \nu(t)\lambda_D(t)b_t^4/a_t^2. \tag{B.7}$$

These identities concern differences in population losses as well as their minimizers; the irreducible conditional-variance terms cancel in comparisons.

Sufficient regularity. For either fitting problem, let ϑ collect all parameters: $(\theta, u_1, \ldots, u_K)$ for joint fitting and (θ, u) for target-only fitting. The empirical minimizer is taken over a compact parameter chart. Its population minimizer ϑ° is unique, interior, and separated: outside every fixed neighborhood, the population objective has a positive gap above its minimum. Near ϑ°, the loss is three times continuously differentiable, its Hessian is positive definite and locally Lipschitz, and integrable derivative envelopes permit differentiation under expectation. The target risk is bounded over the chart and has bounded third derivatives locally.

For N independent original class-c observations, write $\ell_c(\vartheta; x)$ for the relevant loss. Assume

$$\left\| \sup_{\vartheta} \left\| \frac{1}{N} \sum_{i=1}^{N} \nabla_\vartheta^j \ell_c(\vartheta; X_i^c) - \mathbb{E} \nabla_\vartheta^j \ell_c(\vartheta; X_0^c) \right\| \right\|_{L^8} \le C N^{-1/2}, \qquad j = 0, 1, 2. \tag{B.8}$$

For $j = 0$, the inner norm is absolute value and the supremum is over the full chart; for $j = 1, 2$, it is a Euclidean or operator norm and the supremum is needed only in a fixed neighborhood of the minimizer. For fixed K and positive class proportions, these classwise bounds also control the joint criterion at scale $n^{-1/2}$.

Under local alternatives, the conditions hold uniformly, including separation, interiority, and curvature bounds. Population Hessians, observation-gradient covariances, and evaluation Gram matrices converge to their baseline values. For generator-induced scores, the latent law is fixed or has a finite-dimensional parameter included in θ. These are sufficient conditions for expected-risk expansions, stronger than a distributional central limit theorem alone.

Mean-square linearization. Let $L = \mathbb{E}\widehat{L}$, $H = \nabla^2 L(\vartheta^\circ)$, $g_N = \nabla \widehat{L}(\vartheta^\circ)$, and $\Delta = \widehat{\vartheta} - \vartheta^\circ$. Then $\mathbb{E} g_N = 0$. Separation and (B.8) place the minimizer in a fixed interior neighborhood with failure probability $O(N^{-4})$. The same probability bound controls failure of local empirical positive definiteness. On the resulting good event, the averaged empirical Hessian

$$A_N = \int_0^1 \nabla^2 \widehat{L}(\vartheta^\circ + s\Delta)\, ds$$

has a uniformly bounded inverse. The estimating equation gives $\Delta = -A_N^{-1} g_N$, hence $\|\Delta\| \le C \|g_N\|$. Define $R_N = \Delta + H^{-1} g_N$ on the full sample space. On the good event,

$$R_N = H^{-1}(H - A_N)\Delta, \qquad \|R_N\| \le C\{U_N \|g_N\| + \|g_N\|^2\},$$

where U_N is the uniform local Hessian deviation and $\|U_N\|_{L^8} = O(N^{-1/2})$. Compactness controls Δ on the exceptional event, while Hölder's inequality and the eighth-moment bound control g_N. Consequently,

$$\widehat{\vartheta} - \vartheta^\circ = -H^{-1} g_N + R_N, \qquad \mathbb{E}\|R_N\|^2 = O(N^{-2}), \quad \mathbb{E}\|\Delta\|^4 = O(N^{-2}), \quad \|\mathbb{E}\Delta\| = O(N^{-1}). \tag{B.9}$$

For example, on the good event Cauchy–Schwarz gives $\mathbb{E}[U_N^2 \|g_N\|^2] = O(N^{-2})$; on its complement, $\mathbb{E}[\|g_N\|^2 \mathbf{1}_{E_N^c}] \le (\mathbb{E}\|g_N\|^8)^{1/4} \mathbb{P}(E_N^c)^{3/4} = O(N^{-4})$. The same argument is uniform along the stated local alternatives.

B.3 General leading risks

At a correctly specified baseline, define the score Jacobian

$$J_c(x, t) = [\partial_\theta s_{\theta,u}(x, t), \partial_u s_{\theta,u}(x, t)]_{(\theta_0, u_c^0)}$$

and one-original-observation gradient $\psi_c(x) = \nabla_{(\theta,u)} \ell(\theta_0, u_c^0; x)$. Then $\mathbb{E}\psi_c = 0$, and

$$H_c = \int_\delta^T \omega(t) \mathbb{E}[J_c^\top J_c] dt, \quad \Omega_c = \mathrm{Var}\{\psi_c(X_0^c)\}, \quad W_c = \int_\delta^T w(t) \mathbb{E}[J_c^\top J_c] dt. \tag{B.10}$$

The true class densities are held fixed when differentiating. The target-risk Hessian is $2W_*$.

Let P_c select (θ, u_c) from the full joint parameter. Define

$$H_{\mathrm{J}} = \sum_c \rho_c P_c^\top H_c P_c, \qquad \Omega_{\mathrm{J},n} = \sum_c \frac{\rho_c^2}{n_c} P_c^\top \Omega_c P_c, \qquad W_{\mathrm{J}} = P_*^\top W_* P_*. \tag{B.11}$$

The gradient covariance has the factors ρ_c^2/n_c because the independent units are original observations; repeated noise or time draws from the same observation do not create independent original samples. Set

$$V_{\mathrm{J},n} = \operatorname{tr}(W_{\mathrm{J}} H_{\mathrm{J}}^{-1} \Omega_{\mathrm{J},n} H_{\mathrm{J}}^{-1}), \qquad V_{\mathrm{T},M} = \frac{1}{M} \operatorname{tr}(W_* H_*^{-1} \Omega_* H_*^{-1}). \tag{B.12}$$

Lemma 1 (General expected-risk comparison)**.** *Under correct specification and Assumption 3,*

$$\mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{J}}) = V_{\mathrm{J},n} + o(n^{-1}), \qquad \mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{T}}) = V_{\mathrm{T},M} + o(n^{-1}). \tag{B.13}$$

This holds for both sample budgets, arbitrary admissible w, ω, and nonproportional class weights. A leading difference bounded above by $-\gamma/n$ or below by γ/n, for fixed $\gamma > 0$, determines positive or negative transfer, respectively.

Proof. Correct specification makes the target risk and its gradient vanish at the baseline. Locally,

$$\mathcal{R}_*(\vartheta_0 + \Delta) = \Delta^\top W_{\mathrm{J}} \Delta + O(\|\Delta\|^3).$$

By (B.9), $\mathbb{E}\,\|\Delta\|^3 = O(n^{-3/2})$. Bounded risk, compactness, and the $O(n^{-4})$ exceptional probability control the complement of the local event. Write $\Delta = -H_{\mathrm{J}}^{-1} g_n + R_n$, where $\mathbb{E} g_n = 0$ and $\operatorname{Var}(g_n) = \Omega_{\mathrm{J},n}$. The leading quadratic expectation is $V_{\mathrm{J},n}$. Its cross term with R_n is $O(n^{-3/2})$ by Cauchy–Schwarz, and the remainder square is $O(n^{-2})$. The target-only calculation uses covariance Ω_*/M and $M \asymp n$. Subtracting proves the comparison; the $o(n^{-1})$ remainder is eventually smaller than half the assumed margin. No equality $\Omega_c = H_c$ is used. □

B.4 Profiling the class coordinates

We prove Theorem 1. Let $w = \omega$, so $W_c = H_c$, and partition H_c and ψ_c as in Section 3.2. Bars denote class sample averages. The linearized estimating equations are

$$\sum_c \rho_c (A_c \Delta_\theta + F_c \Delta_{u,c}) = -\sum_c \rho_c \overline{\psi}_{\theta,c} + O_{L^2}(n^{-1}), \quad F_c^\top \Delta_\theta + C_c \Delta_{u,c} = -\overline{\psi}_{u,c} + O_{L^2}(n^{-1}).$$

With the quantities in (3.6) and $D_\rho = \sum_c \rho_c D_c$, elimination gives

$$\Delta_\theta = -D_\rho^{-1} \sum_c \rho_c \overline{\zeta}_c + O_{L^2}(n^{-1}), \qquad \Delta_{u,c} = -C_c^{-1} \overline{\psi}_{u,c} - C_c^{-1} F_c^\top \Delta_\theta + O_{L^2}(n^{-1}). \tag{B.14}$$

Complete the square in the target quadratic risk:

$$\begin{pmatrix} \Delta_\theta \\ \Delta_{u,*} \end{pmatrix}^\top H_* \begin{pmatrix} \Delta_\theta \\ \Delta_{u,*} \end{pmatrix} = \Delta_\theta^\top D_* \Delta_\theta + (\Delta_{u,*} + C_*^{-1} F_*^\top \Delta_\theta)^\top C_* (\Delta_{u,*} + C_*^{-1} F_*^\top \Delta_\theta).$$

Applied to the exact leading linear term, this identity separates shared and adjusted class-coordinate contributions even if their observation gradients are correlated. Independence across original observations gives

$$\begin{aligned} V_{\mathrm{J},n} &= \operatorname{tr}\left[D_* D_\rho^{-1} \left(\sum_c \frac{\rho_c^2}{n_c} K_c \right) D_\rho^{-1} \right] + \frac{\operatorname{tr}(C_*^{-1} \Omega_{uu,*})}{n_*}, \\ V_{\mathrm{T},M} &= \frac{\operatorname{tr}(D_*^{-1} K_*) + \operatorname{tr}(C_*^{-1} \Omega_{uu,*})}{M}. \end{aligned} \tag{B.15}$$

For proportional weights, $D_\rho \to D$ and $\sum_c \rho_c^2 K_c / n_c = K/n + o(n^{-1})$. Substitution into Lemma 1 proves (3.8) and (3.9) with (3.7).

B.5 Directional sufficient conditions

Under Theorem 1, equal-total improvement is equivalent at leading order to

$$\mathcal{C}_{\mathrm{T}} - \mathcal{C}_{\mathrm{J}} > (\kappa_*^{-1} - 1)\mathcal{C}_{\mathrm{cl}}. \tag{B.16}$$

Normalize relative to target curvature:

$$M_0 = D_*^{-1/2} D D_*^{-1/2}, \quad L_0 = D_*^{-1/2} K D_*^{-1/2}, \quad B_0 = D_*^{-1/2} K_* D_*^{-1/2}.$$

Then $\mathcal{C}_{\mathrm{J}} = \mathrm{tr}(M_0^{-1} L_0 M_0^{-1})$ and $\mathcal{C}_{\mathrm{T}} = \mathrm{tr}(B_0)$. If $M_0 \succeq \lambda I$ and $L_0 \preceq \nu B_0$, then $\mathcal{C}_{\mathrm{J}} \leq (\nu/\lambda^2)\mathcal{C}_{\mathrm{T}}$, so a sufficient condition is

$$(1 - \nu/\lambda^2)\mathcal{C}_{\mathrm{T}} > (\kappa_*^{-1} - 1)\mathcal{C}_{\mathrm{cl}}.$$

More precisely, let v_j be an orthonormal eigenbasis of M_0 with eigenvalues $\lambda_j > 0$, and set $\ell_j = v_j^\top L_0 v_j$, $b_j = v_j^\top B_0 v_j$. Then

$$\mathcal{C}_{\mathrm{T}} - \mathcal{C}_{\mathrm{J}} = \sum_j \left(b_j - \frac{\ell_j}{\lambda_j^2} \right).$$

If $\ell_j \leq \nu_j b_j$, it suffices that $\sum_j (1 - \nu_j/\lambda_j^2) b_j > (\kappa_*^{-1} - 1)\mathcal{C}_{\mathrm{cl}}$. No simultaneous diagonalization of L_0 and B_0 is required.

For unequal weights, suppose the shared/class training blocks satisfy $F_c = 0$ for every class, and the target evaluation cross-block is zero. Then $D_c = A_c$ and the same budget decomposition holds with

$$\mathcal{C}_{\mathrm{J}} = \mathrm{tr}(W_{\theta\theta,*} D^{-1} K D^{-1}), \quad \mathcal{C}_{\mathrm{T}} = \mathrm{tr}(W_{\theta\theta,*} D_*^{-1} K_* D_*^{-1}), \quad \mathcal{C}_{\mathrm{cl}} = \mathrm{tr}(W_{uu,*} C_*^{-1} \Omega_{uu,*} C_*^{-1}).$$

Set $A_0 = D_*^{-1/2} W_{\theta\theta,*} D_*^{-1/2}$. The shared coefficients become $\mathcal{C}_{\mathrm{J}} = \mathrm{tr}(A_0 M_0^{-1} L_0 M_0^{-1})$ and $\mathcal{C}_{\mathrm{T}} = \mathrm{tr}(A_0 B_0)$. If A_0 and M_0 commute, with common eigenvalues $a_j \geq 0$ and λ_j, a sufficient condition is

$$\sum_j a_j (1 - \nu_j/\lambda_j^2) b_j > (\kappa_*^{-1} - 1)\mathcal{C}_{\mathrm{cl}}.$$

Without commutation, if $a_- I \preceq A_0 \preceq a_+ I$, $M_0 \succeq \lambda I$, and $L_0 \preceq \nu B_0$, it suffices that

$$(a_- - a_+ \nu/\lambda^2)\,\mathrm{tr}(B_0) > (\kappa_*^{-1} - 1)\mathcal{C}_{\mathrm{cl}}.$$

Without the stated orthogonality conditions, use the full variance formulas (B.12).

B.6 Translation families

We prove Corollary 1. Translation equivariance gives $s_{\theta,u}(x,t) = s_{\theta,0}(x - a_t \mathsf{T} u, t)$ and hence

$$\ell(\theta, u_c^0 + v; x + \mathsf{T} u_c^0) = \ell(\theta, v; x).$$

Centered observations have the same law across classes. Differentiating this identity shows that $H_c = H$, $\Omega_c = \Omega$, and all profiled quantities agree. For $w = \omega$, (B.15) yields (3.10), with

$$\mathcal{C}_{\mathrm{sh}} = \mathrm{tr}(D^{-1} K), \qquad \mathcal{C}_{\mathrm{cl}} = \mathrm{tr}(C^{-1} \Omega_{uu}).$$

For proportional weights, the equal-target gap is $\mathcal{C}_{\mathrm{sh}}(1/n - 1/n_*) + o(n^{-1})$, whereas the equal-total gap is $\mathcal{C}_{\mathrm{cl}}(1/n_* - 1/n) + o(n^{-1})$. Strict signs require the corresponding coefficient to be positive; positive Hessians alone do not imply positive gradient variances.

For unequal weights, assume the symmetry in the corollary. The centered noised density is even and its score is odd. Its shape derivatives are odd, whereas $\partial_u s_{\theta,u}|_{u=0} = -a_t J_x s_{\theta,0} \mathsf{T}$ is even. Thus their cross-products integrate to zero at each time, under both weights. The training and evaluation matrices are block diagonal, giving

$$\mathcal{C}_{\mathrm{sh}} = \mathrm{tr}(W_\theta A^{-1} \Omega_{\theta\theta} A^{-1}), \qquad \mathcal{C}_{\mathrm{cl}} = \mathrm{tr}(W_u C^{-1} \Omega_{uu} C^{-1}).$$

The same risk decomposition follows from (B.12), without requiring $\Omega_{\theta u} = 0$.

B.7 The geometric sensitivity construction

We prove Proposition 2. The first generator coordinate is z, so the map is injective and its derivative has norm at least one. Its second derivative is

$$G''_{\theta,u}(z) = (0, 2\lambda u\theta_1, \ldots, 2\lambda u\theta_4, 0).$$

For $\lambda > 0$, $u > 0$, and $\theta \neq 0$, this vector is nonzero and cannot be parallel to the first derivative, whose first coordinate is one. Its nonzero normal component gives a curved embedded interval. On a compact chart with positive class coordinates, the last-coordinate mean identifies u, and the means of coordinates $2, \ldots, 5$ identify θ through $u\theta_j(1 + \lambda/3)$.

The true generators also admit the centered form in Assumption 2. Let $\overline{u} = \sum_c \pi_c u_c^0$, $G_0 = G_{\theta_0,\overline{u}}$, and

$$D(z) = (0, \theta_{0,1} f_\lambda(z), \ldots, \theta_{0,4} f_\lambda(z), 1)^\top.$$

Then $G_c = G_0 + D(u_c^0 - \overline{u})$, with centered class coefficients. This is a representation of the true generators, not a constraint supplied to fitting.

At $(\lambda, \theta_0) = (0, 0)$, the first observation coordinate carries no fitted parameter. The remaining five coordinates are independent Gaussian variables with variance τ^2 and candidate mean $(u\theta_1, \ldots, u\theta_4, u)$. Define

$$h = \int_\delta^T \omega(t) \frac{a_t^2}{\sigma_t^4}\, dt, \qquad C = \tau^2 \int_\delta^T w(t) \frac{a_t^2}{\sigma_t^4}\, dt.$$

For one Gaussian coordinate with candidate mean μ, differentiation of (3.2) gives $-h(x_{0,k} - \mu)$. At the true parameters,

$$\partial_{\theta_j} \ell = -h u_c^0 X_{0,j+1}^c, \qquad \partial_{u_c} \ell = -h(X_{0,6}^c - u_c^0).$$

For $(u_1^0, u_2^0, u_3^0) = (1, 2, 2)$, parameter order (θ, u_1, u_2, u_3) gives

$$H_{\mathrm{J}} = h\,\mathrm{diag}(3I_4, \tfrac{1}{3}I_3), \qquad \Omega_{\mathrm{J},n} = \frac{\tau^2 h}{n} H_{\mathrm{J}}, \qquad W_{\mathrm{J}} = \frac{C}{\tau^2}\,\mathrm{diag}(I_4, 1, 0, 0).$$

Thus the leading parameter covariance is $(\tau^2/n)\,\mathrm{diag}(\frac{1}{3}I_4, 3I_3)$, and

$$V_{\mathrm{J},n} = \frac{C}{n}(4/3 + 3) = \frac{13C}{3n}.$$

For target-only fitting, $H_1 = hI_5$, $\Omega_1 = \tau^2 h^2 I_5$, and $W_1 = (C/\tau^2)I_5$, giving $V_{\mathrm{T},n} = 5C/n$. Lemma 1 proves (3.11), with a strict gap $-2C/(3n) + o(n^{-1})$.

Replacing the true coordinates by $(r, 2r, 2r)$ multiplies the shared covariance block by r^{-2} and the shared evaluation block by r^2; these factors cancel. The class-coordinate contribution is unchanged, while the centered deformation is $O(r)$ in C^2. For each fixed $r > 0$, bounded generator derivatives and the positive noise lower bound give common polynomial envelopes near $(\lambda, \theta_0) = (0, 0)$. Dominated convergence makes H_c, Ω_c, and W_c continuous there, and positive definiteness persists locally. The negative leading risk gap therefore persists in a neighborhood, including curved examples. The regular fitting conditions are retained as assumed in the proposition.

B.8 The quadratic score-direction construction

We prove Theorem 2. Write $v_t = a_t^2\lambda_0 + b_t^2$ and $U = X_0^c - \mu_c \sim \mathcal{N}(0, \lambda_0)$. At the true parameters, the score Jacobian in order (λ, β, u_c) is

$$J_c(x, t) = \left(\frac{a_t^2(x - a_t\mu_c)}{v_t^2},\ f(t)\{x^2 - a_t^2\mu_c^2 - v_t\},\ \frac{a_t}{v_t} \right).$$

Define

$$A_\omega = \int_\delta^T \omega(t)\frac{a_t^4}{v_t^3}dt, \qquad B_\omega = \int_\delta^T \omega(t)\frac{a_t^2}{v_t^2}dt,$$
$$F_\omega = \int_\delta^T \omega(t)\frac{f(t)a_t^3}{v_t}dt, \qquad G_\omega = \int_\delta^T \omega(t)f(t)^2 v_t^2 dt,$$
$$L_\omega = \int_\delta^T \omega(t)f(t)^2 a_t^2 v_t dt.$$

Use A_w, B_w, G_w for the corresponding evaluation integrals. Gaussian moments give

$$H_c = \begin{pmatrix} A_\omega & 2\mu_c F_\omega & 0 \\ 2\mu_c F_\omega & 2G_\omega + 4\mu_c^2 L_\omega & 0 \\ 0 & 0 & B_\omega \end{pmatrix}. \tag{B.17}$$

For example, with $Y = X_t^c - a_t\mu_c$, $\mathbb{E}Y^2 = v_t$, $\mathbb{E}[Y\{(X_t^c)^2 - a_t^2\mu_c^2 - v_t\}] = 2a_t\mu_c v_t$, and $\mathbb{E}\{(X_t^c)^2 - a_t^2\mu_c^2 - v_t\}^2 = 2v_t^2 + 4a_t^2\mu_c^2 v_t$.

Let $H_2(U) = U^2 - \lambda_0$ and $H_3(U) = U^3 - 3\lambda_0 U$. The true DSM residual is

$$s_c(X_t^c, t) + \varepsilon/b_t = -a_t U/v_t + a_t^2\lambda_0\varepsilon/(b_t v_t).$$

Differentiation of the original-observation loss and integration over corruption noise yield

$$\psi_c(U) = -\begin{pmatrix} A_\omega H_2(U) \\ F_\omega\{H_3(U) + 2\mu_c H_2(U)\} \\ B_\omega U \end{pmatrix}. \tag{B.18}$$

The same original observation enters every time integral; consequently, its gradient covariance contains squares of time integrals, not integrals of squared independent gradients. Hermite orthogonality gives

$$\Omega_c = \begin{pmatrix} 2A_\omega^2\lambda_0^2 & 4\mu_c A_\omega F_\omega \lambda_0^2 & 0 \\ 4\mu_c A_\omega F_\omega \lambda_0^2 & F_\omega^2(6\lambda_0^3 + 8\mu_c^2\lambda_0^2) & 0 \\ 0 & 0 & B_\omega^2\lambda_0 \end{pmatrix}. \tag{B.19}$$

The Hessian is positive definite: $A_\omega, B_\omega, G_\omega, L_\omega > 0$, $F_\omega^2 \le A_\omega L_\omega$, and the shared-block determinant is $2A_\omega G_\omega + 4\mu_c^2(A_\omega L_\omega - F_\omega^2) > 0$. Zero population score error identifies $\beta = 0$ from its quadratic coefficient, then $\lambda = \lambda_0$ from its linear coefficient, and $u_c = \mu_c$ from its constant coefficient. Compactness gives separation. After noise integration, the loss and its required derivatives are polynomials of bounded degree in the original Gaussian observation, with coefficients uniformly bounded on the chart. Their empirical deviations reduce to finitely many Gaussian sample moments, giving (B.8). Thus Lemma 1 applies.

The shared/class blocks vanish. Set $q_\mu = \sum_c \kappa_c\mu_c^2$. Under $\sum_c \kappa_c\mu_c = 0$, the limiting profiled matrices are

$$D = \mathrm{diag}(A_\omega, 2G_\omega + 4q_\mu L_\omega), \quad K = \mathrm{diag}\{2A_\omega^2\lambda_0^2, F_\omega^2(6\lambda_0^3 + 8q_\mu\lambda_0^2)\}.$$

The leading joint shared covariance is

$$\frac{1}{n}D^{-1}KD^{-1} = \frac{1}{n}\,\mathrm{diag}\left\{2\lambda_0^2, \frac{F_\omega^2\lambda_0^2(3\lambda_0 + 4q_\mu)}{2(G_\omega + 2q_\mu L_\omega)^2}\right\}.$$

The target-coordinate variance is λ_0/n_* and its leading covariance with the shared parameters is zero. Since $\mu_* = 0$, $W_* = \mathrm{diag}(A_w, 2G_w, B_w)$. Hence the coefficients in (3.13) are

$$C_\lambda = 2A_w\lambda_0^2, \qquad \mathcal{C}_{\mathrm{cl}} = B_w\lambda_0,$$
$$C_{\beta,\mathrm{T}} = \frac{3G_w F_\omega^2\lambda_0^3}{G_\omega^2}, \qquad C_{\beta,\mathrm{J}} = \frac{G_w F_\omega^2\lambda_0^2(3\lambda_0 + 4q_\mu)}{(G_\omega + 2q_\mu L_\omega)^2}. \tag{B.20}$$

The trace formula proves the theorem and its transfer condition. Finally, $C_{\beta,\mathrm{J}} \to 0$ as $q_\mu \to \infty$. With two auxiliary classes, their means can be scaled in opposite weighted proportions to retain zero weighted mean. If the stated strict condition on $C_{\beta,\mathrm{T}}$ holds, a sufficiently large but finite configuration gives the required margin. Its means and compact chart remain fixed while the sample size grows.

B.9 Fixed incompatibility

Let $\vartheta_{\mathrm{J}}^{\circ}$ minimize the limiting joint objective $\mathcal{L}_{\mathrm{J},\infty}(\vartheta) = \sum_c \overline{\rho}_c \mathcal{L}_c(\theta, u_c)$, and let s_{J} be its target predictor.

For (3.15), the jointly representable fields $h_0 + H_0 u_c$ are a feasible competitor, with class score errors $-r_c$. By population optimality and (B.5),

$$\sum_c \overline{\rho}_c \int_\delta^T \omega(t) \left\| s_{\theta_{\mathrm{J}}^{\circ}, u_{c,\mathrm{J}}^{\circ}} - s_c \right\|^2_{L^2(p_t^c)} dt \le \sum_c \overline{\rho}_c \int_\delta^T \omega(t) \left\| r_c \right\|^2_{L^2(p_t^c)} dt \le C_{\delta,\omega}(\epsilon^4 + \eta^2).$$

Nonnegativity of every class term and $w \le L_w \omega$ prove the target bound after division by $\overline{\rho}_*$.

The uniform loss law in (B.8), positive limiting weights, compactness, and population separation imply consistency of the joint empirical minimizer for $\vartheta_{\mathrm{J}}^{\circ}$. The target-only minimizer converges to its correctly specified parameter. Continuity and boundedness of the target risk then give convergence in expectation, proving (3.16).

For the sufficient condition, let $u_c^{\circ}(\theta)$ be the smooth unique classwise optimizer near θ_0. Its first-order condition gives the envelope identity

$$\nabla_\theta \inf_u \mathcal{L}_c(\theta, u) = \partial_\theta \mathcal{L}_c(\theta, u_c^{\circ}(\theta)).$$

Condition (3.17) rules out θ_0 as an interior minimizer of the profiled joint objective, so $\theta_{\mathrm{J}}^{\circ} \neq \theta_0$. The stated target-identification condition then forces $\mathcal{R}_*(s_{\mathrm{J}}) > 0$, proving the fixed-mismatch part of Theorem 3.

B.10 Local geometric bias

Let $\mathbb{E}_n$ denote expectation under the perturbations (3.18). The smoothing bounds give, uniformly over time and classes,

$$s_{G_{c,n}}(x,t) = s_c(x,t) + h_n \dot{s}_{G_c}[V_c](x,t) + O\{h_n^2(1 + \|x\|^2)\}. \tag{B.21}$$

At the baseline fitted parameter, differentiate the observation loss and use (B.4) under the perturbed data law:

$$\mathbb{E}_n \psi_c(X_{0,n}^c) = \int_\delta^T \omega(t) \mathbb{E}_n[J_c^\top (s_c - s_{G_{c,n}})] dt = -h_n b_c + o(h_n),$$

where

$$b_c = \int_\delta^T \omega(t) \mathbb{E}_{p_t^c}[J_c(X_t,t)^\top \dot{s}_{G_c}[V_c](X_t,t)] dt, \qquad b_* = 0. \tag{B.22}$$

To justify passage to the baseline expectation, couple the original and perturbed observations through the same (Z, ξ, ε). Their noised difference is $a_t h_n V_c(Z) \to 0$; the uniform integrable envelopes in Assumption 3 permit dominated convergence.

Set $b_\rho = \sum_c \rho_c P_c^\top b_c$. The joint population gradient at ϑ_0 is $-h_n b_\rho + o(h_n)$. Uniform separation places the perturbed minimizer $\vartheta_{\mathrm{J},n}^{\circ}$ near ϑ_0. Expanding its zero gradient and using the uniformly invertible Hessian gives

$$\vartheta_{\mathrm{J},n}^{\circ} - \vartheta_0 = h_n H_{\mathrm{J}}^{-1} b_\rho + o(h_n). \tag{B.23}$$

The target law is unchanged and correctly fitted at ϑ_0. Its local quadratic expansion therefore yields

$$\mathcal{R}_*(s_{\mathrm{J},n}) = h_n^2 b_\rho^\top H_{\mathrm{J}}^{-1} W_{\mathrm{J}} H_{\mathrm{J}}^{-1} b_\rho + o(h_n^2). \tag{B.24}$$

For the empirical fit, write $\Delta_n = \widehat{\vartheta}_{\mathrm{J}} - \vartheta^\circ_{\mathrm{J},n}$. Uniform mean-square linearization gives $\|\mathbb{E}\Delta_n\| = O(n^{-1})$, $\mathbb{E}\,\|\Delta_n\|^2 = O(n^{-1})$, and $\mathbb{E}\,\|\Delta_n\|^4 = O(n^{-2})$. The target-risk gradient at the population minimizer is $O(|h_n|)$; it need not vanish merely because the joint DSM gradient does. Its expected linear contribution is therefore $O(|h_n|/n) = O(n^{-3/2})$. The target-risk Hessian differs from $2W_{\mathrm{J}}$ by $O(|h_n|)$, so its replacement in the quadratic term has the same negligible order. The cubic remainder is $O(n^{-3/2})$, and the exceptional event is controlled as in Lemma 1. Continuity of the population matrices then gives

$$\mathbb{E}\mathcal{R}_*(\widehat{s}_{\mathrm{J}}) = \mathcal{R}_*(s_{\mathrm{J},n}) + V_{\mathrm{J},n} + o(n^{-1}).$$

The unchanged target-only model has risk $V_{\mathrm{T},M} + o(n^{-1})$. Subtraction proves (3.19) and the strict-margin comparisons.

Profiling the local bias. For $w = \omega$, partition $b_c = (b_{\theta,c}^\top, b_{u,c}^\top)^\top$. Eliminating class-coordinate changes from (B.23) gives

$$\theta^\circ_{\mathrm{J},n} - \theta_0 = h_n D_\rho^{-1} \sum_c \rho_c (b_{\theta,c} - F_c C_c^{-1} b_{u,c}) + o(h_n). \tag{B.25}$$

Because $b_* = 0$, the target coordinate changes by $-C_*^{-1}F_*^\top(\theta^\circ_{\mathrm{J},n} - \theta_0) + o(h_n)$. Completing the target square thus gives

$$\mathcal{R}_*(s_{\mathrm{J},n}) = (\theta^\circ_{\mathrm{J},n} - \theta_0)^\top D_* (\theta^\circ_{\mathrm{J},n} - \theta_0) + o(h_n^2).$$

If a class deformation can be represented to first order by changing its own coordinate, so $\dot{s}_{G_c}[V_c] = \partial_u s_{\theta_0,u_c^0} d_c$, then $b_{\theta,c} = F_c d_c$ and $b_{u,c} = C_c d_c$. Its profiled forcing is zero. More generally, nonzero class forcings can cancel in their weighted sum; such cancellation removes the first-order shared shift.

For the translation family with class-specific shape changes $\theta_0 + h_n a_c$, the score change is $\partial_\theta s_{\theta_0,u_c^0} a_c$. Hence $b_{\theta,c} = A_c a_c$, $b_{u,c} = F_c^\top a_c$, and the profiled forcing is $D_c a_c$. Translation equivariance makes $D_c = D$ for all classes, so $\theta^\circ_{\mathrm{J},n} - \theta_0 = h_n \sum_c \rho_c a_c + o(h_n)$. This proves (3.20). Under the symmetry conditions of Corollary 1, unequal weights replace D in that risk formula by the target evaluation block W_θ.

B.11 An approximation mechanism with unequal weights

The following misspecified family shows that target score risk and target DSM loss can have opposite transfer signs when their noise-level weights differ. It is separate from the correctly specified comparisons above.

Use a finite set of noise levels, with positive normalized evaluation weights w_t and training weights ω_t. Let

$$X_0^c = Y + \mu_c, \qquad Y = G_0(Z) + \tau\xi, \qquad \mathbb{E}Y = 0, \qquad \lambda = d^{-1}\mathbb{E}\,\|Y\|^2 > 0, \qquad \mu_* = 0,$$

where G_0 is a bounded smooth embedding of a compact latent domain. Write $X_t^c = a_t X_0^c + b_t \varepsilon$, $v_t = a_t^2\lambda + b_t^2$, with $a_t, b_t > 0$, and fit the time-independent family

$$s_{\theta,u_c}(x,t) = -\theta x + u_c, \qquad \theta \in \mathbb{R}, \quad u_c \in \mathbb{R}^d.$$

For fixed positive class weights, define

$$V_\omega = \sum_t \omega_t v_t, \quad V_w = \sum_t w_t v_t, \quad A_\omega = \sum_t \omega_t a_t, \quad S_\omega = \sum_t \omega_t (a_t - A_\omega)^2, \quad Q_\mu = \frac{1}{d}\sum_c \rho_c \,\|\mu_c\|^2 .$$

Proposition 3 (Noise-level approximation)**.** *The target-only and joint population DSM minimizers satisfy*

$$\theta_{\mathrm{T}}^{\circ} = V_{\omega}^{-1}, \quad u_{*,\mathrm{T}}^{\circ} = 0, \qquad \theta_{\mathrm{J}}^{\circ} = (V_{\omega} + Q_{\mu} S_{\omega})^{-1}, \quad u_{c,\mathrm{J}}^{\circ} = \theta_{\mathrm{J}}^{\circ} A_{\omega} \mu_c. \tag{B.26}$$

Let $s_{\mathrm{T}} = s_{\theta_{\mathrm{T}}^{\circ},0}$ *and* $s_{\mathrm{J}} = s_{\theta_{\mathrm{J}}^{\circ},0}$*. If*

$$S_{\omega} > 0, \qquad V_{\omega} < V_w, \qquad 0 < Q_{\mu} \le (V_w - V_{\omega})/S_{\omega},$$

then

$$\begin{aligned} \mathcal{R}_*(s_{\mathrm{J}}) - \mathcal{R}_*(s_{\mathrm{T}}) &= d V_w (\theta_{\mathrm{J}}^{\circ} - \theta_{\mathrm{T}}^{\circ})(\theta_{\mathrm{J}}^{\circ} + \theta_{\mathrm{T}}^{\circ} - 2/V_w) < 0, \\ \mathcal{L}_*(\theta_{\mathrm{J}}^{\circ}, 0) - \mathcal{L}_*(\theta_{\mathrm{T}}^{\circ}, 0) &= \tfrac{1}{2} d V_{\omega} (\theta_{\mathrm{J}}^{\circ} - \theta_{\mathrm{T}}^{\circ})^2 > 0. \end{aligned} \tag{B.27}$$

The same strict signs hold for expected population risks and losses of empirical DSM fits under sufficiently large equal-total budgets, with positive limiting class proportions. For a variance-preserving schedule, the conditions are attainable when $\lambda > 1$*,* a_t *is nonconstant, and* $\omega_t = w_t b_t^2 / \sum_r w_r b_r^2$*. With these weights, joint fitting also increases target noise-prediction MSE under noise-level sampling weights* w_t*.*

Proof. Identity (B.4) gives $\mathbb{E} s_c(X_t^c, t) = 0$ and $\mathbb{E}[(X_t^c)^{\top} s_c(X_t^c, t)] = -d$. Consequently,

$$\mathbb{E} \left\| s_{\theta, u_c}(X_t^c, t) - s_c(X_t^c, t) \right\|^2 = d(v_t \theta^2 - 2\theta) + \left\| u_c - \theta a_t \mu_c \right\|^2 + \mathbb{E} \left\| s_c(X_t^c, t) \right\|^2 .$$

Completing the intercept square gives

$$\sum_t \omega_t \left\| u_c - \theta a_t \mu_c \right\|^2 = \left\| u_c - \theta A_{\omega} \mu_c \right\|^2 + \theta^2 S_{\omega} \left\| \mu_c \right\|^2 .$$

The profiled joint objective is therefore $\frac{d}{2}\{(V_{\omega} + Q_{\mu} S_{\omega})\theta^2 - 2\theta\} + \text{constant}$. The target-only expression sets $Q_{\mu} = 0$, proving (B.26).

For the target, $\mathcal{R}_*(s_{\theta,u}) = d V_w (\theta - V_w^{-1})^2 + \|u\|^2 + \text{constant}$, while $\mathcal{L}_*(\theta, u) = \frac{1}{2} d V_{\omega} (\theta - V_{\omega}^{-1})^2 + \frac{1}{2} \|u\|^2 + \text{constant}$. The assumptions imply $V_w^{-1} \le \theta_{\mathrm{J}}^{\circ} < \theta_{\mathrm{T}}^{\circ} = V_{\omega}^{-1}$ and hence the strict signs.

For the empirical joint fit, eliminating intercepts gives $\widehat{u}_{c,\mathrm{J}} = A_{\omega} \widehat{\theta}_{\mathrm{J}} \overline{X}_c$ and

$$\widehat{\theta}_{\mathrm{J}}^{-1} = \sum_t \omega_t b_t^2 + \frac{A_{\omega}^2 + S_{\omega}}{d} \sum_c \frac{\rho_c}{n_c} \sum_i \left\| X_i^c - \overline{X}_c \right\|^2 + \frac{S_{\omega}}{d} \sum_c \rho_c \left\| \overline{X}_c \right\|^2 .$$

The target-only fit uses one class and n observations. The laws of large numbers give convergence to the population slopes. Both fitted slopes are bounded by $(\sum_t \omega_t b_t^2)^{-1}$, while the target intercept second moment is $O(n_*^{-1})$ or $O(n^{-1})$. The expected quadratic risks and losses therefore converge to their population values. Their strict signs persist. For converging class weights, use their limiting values in Q_{μ}.

For the final claim, $a_t^2 + b_t^2 = 1$ and $\omega_t \propto w_t b_t^2$ give

$$\sum_t \omega_t a_t^2 = \sum_t w_t a_t^2 - \frac{\mathrm{Var}_{t \sim w}(a_t^2)}{\sum_t w_t b_t^2} < \sum_t w_t a_t^2 .$$

Since $v_t = 1 + (\lambda - 1) a_t^2$, this proves $V_{\omega} < V_w$ when $\lambda > 1$; nonconstant a_t also gives $S_{\omega} > 0$. Small nonzero auxiliary translations can meet the bound on Q_{μ}. Finally,

$$\mathbb{E}_{t \sim w, X_0^*, \varepsilon} \left\| \varepsilon - \widehat{\varepsilon}_{\theta,u}(X_t^*, t) \right\|^2 = 2 \left(\sum_t w_t b_t^2 \right) \mathcal{L}_*(\theta, u),$$

so the target noise-MSE ordering follows from the second line of (B.27). □

B.12 Noise domains and parameterizations

Beyond a compact noise interval, the proofs require integrability of the squared envelope in (B.3), of the loss and derivative envelopes, and of the corresponding evaluation terms, uniformly along local alternatives. The explicit Gaussian-noise term is integrable if $\int \omega(t)/b_t^2\, dt < \infty$; this condition alone does not replace the other envelope requirements.

For discrete DDPM noise levels, $a_t = \sqrt{\overline{\alpha}_t}$, $b_t = \sqrt{1-\overline{\alpha}_t}$, and sums replace integrals. Equation (B.6) gives the training weight for noise prediction. For EDM additive-noise coordinates, $a_t = 1$, $b_t = t$, and

$$\lambda_D(t) = \frac{t^2 + \sigma_{\text{data}}^2}{t^2 \sigma_{\text{data}}^2}, \qquad \omega(t) = \nu(t)\left(t^2 + \frac{t^4}{\sigma_{\text{data}}^2}\right). \tag{B.28}$$

If ν is lognormal, every real moment is finite:

$$\int_0^\infty \omega(t) t^r\, dt = \mathbb{E}_\nu\left[t^{r+2} + \frac{t^{r+4}}{\sigma_{\text{data}}^2}\right] < \infty, \qquad r \in \mathbb{R}.$$

For bounded generators with uniformly bounded parameter derivatives through the required orders, Gaussian smoothing gives polynomial observation envelopes involving powers of t and t^{-1}. Gaussian observation moments and the displayed identity control their training-weighted integrals. With the analogous evaluation integrability and statistical regularity, the same arguments extend to this noise domain. The neural experiments use these DSM parameterizations; the weight conversions alone do not establish their parametric regularity assumptions.

C Simulation details

C.1 AR(1) data generation

We use $K = 3$ classes in $d = 100$, with target label 1 and target mean zero. Auxiliary means have the form

$$\mu_k = 20 m_k / \|m_k\|, \qquad m_k \sim \mathcal{N}(0, I_d), \quad k \neq 1.$$

A candidate mean is accepted only if its distance from every previously accepted mean is at least 15. The covariances satisfy $(\Sigma_k)_{ij} = \rho_k^{|i-j|}$. Shared-covariance settings use $\rho_k \in \{0.2, 0, -0.2\}$, common to all classes. In the class-varying setting, the target has $\rho_1 = 0.1$. At zero mismatch, both auxiliary correlations are also 0.1. At nonzero mismatch, one auxiliary correlation is drawn below 0.1 and the other above it, then randomly assigned to the two labels:

Mild:	$\rho_{\text{low}} \sim \text{Unif}(-0.10, 0.00)$,	$\rho_{\text{high}} \sim \text{Unif}(0.20, 0.30)$,
Medium:	$\rho_{\text{low}} \sim \text{Unif}(-0.30, -0.10)$,	$\rho_{\text{high}} \sim \text{Unif}(0.30, 0.50)$,
Strong:	$\rho_{\text{low}} \sim \text{Unif}(-0.55, -0.30)$,	$\rho_{\text{high}} \sim \text{Unif}(0.50, 0.75)$.

Changing these correlations changes both covariance orientation and spectrum. The rotation experiment instead isolates orientation at a fixed spectrum.

C.2 Rotation data generation and pairing

Let $\Lambda = \text{diag}(1.8 I_{50}, 0.2 I_{50})$. Within each seed, a QR decomposition of a standard Gaussian matrix gives an orthogonal matrix U, shared across all angles. Define

$$R_\theta = \begin{pmatrix} \cos\theta I_{50} & -\sin\theta I_{50} \\ \sin\theta I_{50} & \cos\theta I_{50} \end{pmatrix}.$$

The class covariances are

$$\Sigma_1 = U\Lambda U^\top, \qquad \Sigma_2 = (UR_\theta)\Lambda(UR_\theta)^\top, \qquad \Sigma_3 = (UR_{-\theta})\Lambda(UR_{-\theta})^\top.$$

All means are zero. Rotation acts on 50 high/low variance pairs, preserving each covariance's eigenvalues, trace, and determinant. Joint sizes are $n_* \in \{200, 600, 1800, 5400\}$ per class; target-only sizes are $M \in \{200, 600, 1800, 5400, 16200\}$. Target training sets are nested within seed and shared by comparisons with the same target size. Independent streams generate auxiliary and held-out samples, while the same underlying Gaussian draws are reused across angles.

Within each seed and capacity, the same initial conditional parameter vector is copied to C_{T} and all joint models. The unconditional model has a separate parameterization. Each target-only model is trained once per seed, capacity, and sample size, then reused in applicable comparisons. Standard capacity uses time-embedding dimension 64, hidden width 256, four hidden layers, and class-embedding dimension 32. Limited capacity uses time-embedding dimension 64, hidden width 128, two hidden layers, and class-embedding dimension 8. All other settings are common. Each capacity–angle configuration has 20 paired seeds.

C.3 Diffusion and optimization

All Gaussian models use 1,000 DDPM steps,

$$x_t = \sqrt{\overline{\alpha}_t}x_0 + \sqrt{1-\overline{\alpha}_t}\,\varepsilon, \quad \alpha_t = 1-\beta_t, \quad \overline{\alpha}_t = \prod_{s=1}^{t}\alpha_s,$$

with a linear variance schedule from $\beta_1 = 10^{-4}$ to $\beta_{1000} = 2\times 10^{-2}$. It gives $\sqrt{\overline{\alpha}_{1000}} \approx 0.00635$. Sampling starts from a standard Gaussian prior.

Models use SiLU activations and Adam with learning rate 2×10^{-4}, batch size 512, and 20,000 updates. The AR(1) experiment uses the standard architecture. Both conditional controls retain the full three-class embedding table. The target-only conditional model uses the fixed target label during training; the joint model samples class labels uniformly and observations with replacement within class. Both use the target label at evaluation and generation. The unconditional model omits the class input.

C.4 Metrics

The target noised marginal is

$$p_t^* = \mathcal{N}\{\sqrt{\overline{\alpha}_t}\mu_*, \overline{\alpha}_t\Sigma_* + (1-\overline{\alpha}_t)I_d\},$$

so its exact score is available. The reported score risk averages squared Euclidean score error uniformly over diffusion steps. Noise-prediction MSE also averages over steps, but divides squared error by d. The score estimate is $\widehat{s} = -\widehat{\varepsilon}/\sqrt{1-\overline{\alpha}_t}$.

For generated data, we compare the fitted Gaussian $\mathcal{N}(\widehat{m}_{\mathrm{gen}}, \widehat{C}_{\mathrm{gen}})$ with the Gaussian target:

$$W^2_{2,\mathrm{Gauss}} = \|\widehat{m}_{\mathrm{gen}} - \mu_*\|^2 + \mathrm{tr}\Big[\widehat{C}_{\mathrm{gen}} + \Sigma_* - 2\{\Sigma_*^{1/2}\widehat{C}_{\mathrm{gen}}\Sigma_*^{1/2}\}^{1/2}\Big]. \tag{C.1}$$

This measures discrepancies in generated first and second moments; it does not detect every distributional difference if generated samples are non-Gaussian.

Final rotation score risk and noise MSE use 100 independent target observation–noise pairs at each of 1,000 steps, or 100,000 pairs per model. Evaluation is independent of training and validation data and uses common draws within seed. Generated-moment metrics use 5,000 target samples per model from the ancestral DDPM sampler with posterior variance. Noise-level diagnostics group steps 1–100, 101–500, and 501–1000 as low, middle, and high noise.

C.5 Replication and aggregation

Gaussian summaries use 20 independent seeds, with means and standard errors across seeds. For the AR(1) sample-size tables, the four class-varying levels are first averaged within each seed, model, and sample size. This summary and the three shared-covariance settings then receive equal weight. Covariance-level tables instead average within seed over the three sample sizes. In the class-varying regime, the four target-only evaluations are averaged into a common baseline because their training distributions do not depend on auxiliary covariances.

C.6 Paired comparisons

For each sample size, budget, capacity, and metric, differences D_s are formed within the $S = 20$ seeds: $C_{\mathrm{J}} - C_{\mathrm{T}}$, $C_{\mathrm{J}} - U_{\mathrm{T}}$, or $C_{\mathrm{T}} - U_{\mathrm{T}}$. The mean difference and pointwise 95% interval are

$$\widehat{\Delta} = \frac{1}{S}\sum_{s=1}^{S} D_s, \qquad \widehat{\Delta} \pm t_{S-1,0.975}\frac{\widehat{\mathrm{sd}}(D)}{\sqrt{S}}.$$

Negative joint-minus-control differences favor joint training. These are pointwise intervals, without a simultaneous adjustment over the experiment grid.

D Additional rotation results

Figures 2 and 3 give the complete grid for both capacities. Under equal target size, the paired $C_{\mathrm{J}} - C_{\mathrm{T}}$ score-risk intervals are below zero at zero rotation throughout the grid. For each nonzero angle, both joint-minus-control intervals are above zero at $n_* = 1800$ and 5400, under both capacities. Under equal total size, both are above zero for every tested sample size and nonzero angle.

At zero rotation under equal total size, C_{J} improves on C_{T} at $n_* = 600, 1800, 5400$ under both capacities, while U_{T} has lower risk throughout. At $n_* = 5400$, standard-capacity risks are 11.64, 12.41, and 11.09 for C_{J}, C_{T}, and U_{T}. Under equal target size at the same size and limited capacity, $C_{\mathrm{J}} - U_{\mathrm{T}} = 1.51$ with 95% interval $[0.90, 2.13]$, whereas $C_{\mathrm{J}} - C_{\mathrm{T}} = -3.00$ with interval $[-3.46, -2.54]$.

D.1 Noise weighting and metric rankings

For a fixed predictor E, let r_t^E be its population target score risk and m_t^E its per-coordinate noise MSE at step t. Conditional-expectation orthogonality gives, for either control $B \in \{C_{\mathrm{T}}, U_{\mathrm{T}}\}$,

$$m_t^B - m_t^{C_{\mathrm{J}}} = \frac{1 - \overline{\alpha}_t}{d}(r_t^B - r_t^{C_{\mathrm{J}}}). \tag{D.1}$$

Thus the aggregate criteria weight score errors differently. At 30° in Table 3, joint score risk is lower than C_{T} (49.72 versus 65.42), but its noise MSE is higher (0.26825 versus 0.25860). This is consistent with (D.1) when gains and losses occur at different noise levels.

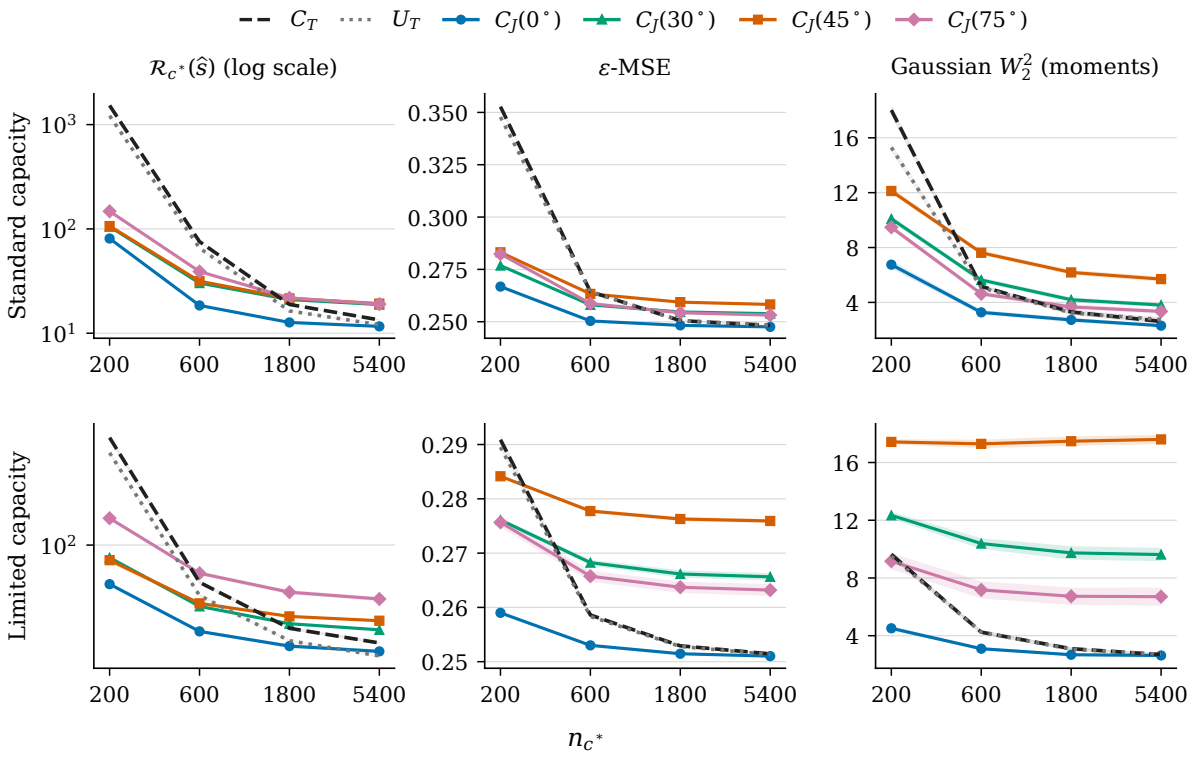


Figure 2: Complete equal-target rotation results. The horizontal axis is n_* and target-only size is $M = n_*$. Rows show standard and limited capacities; columns show score risk, per-coordinate noise MSE, and moment-matched Gaussian W_2^2. Curves are means over 20 seeds and bands are pointwise 95% Student-t intervals. Score-risk and horizontal axes are logarithmic. Lower values are better.

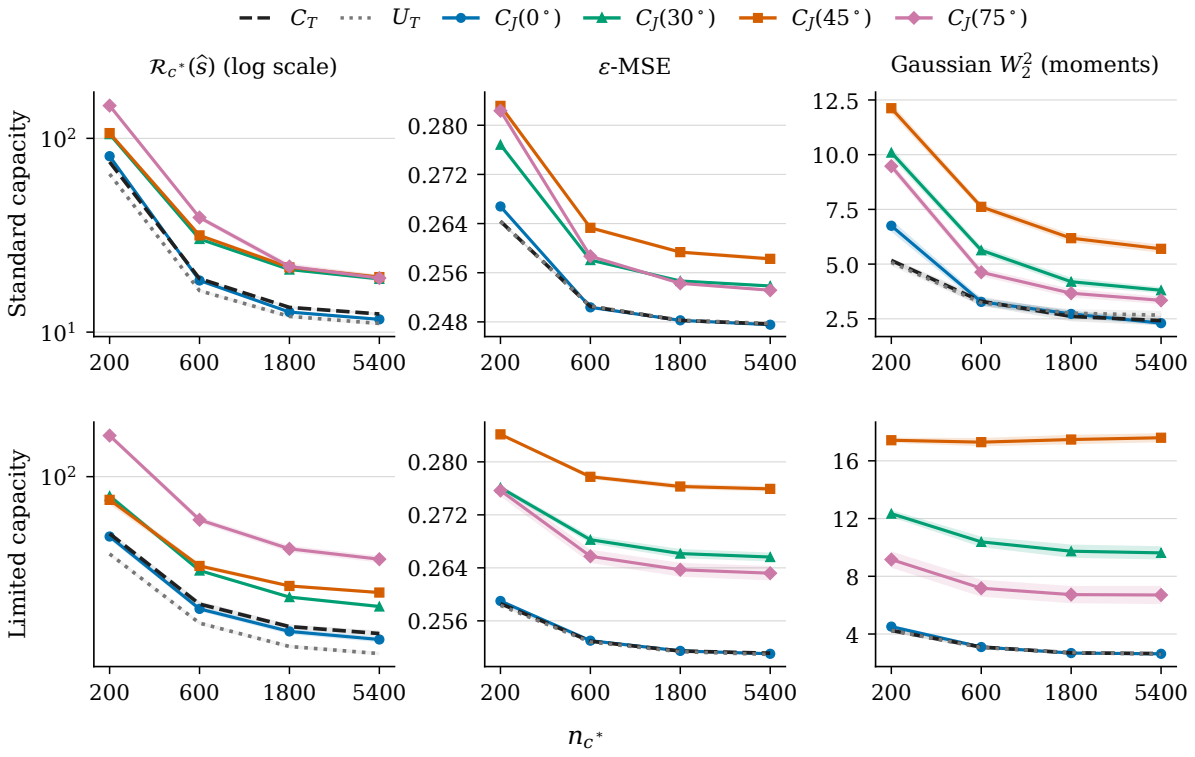


Figure 3: Complete equal-total rotation results. The horizontal axis is n_*, with $M = n = 3n_*$ for every model. Rows, columns, and intervals are defined as in Figure 2.

E Gaussian classes with AR(1) covariances

This experiment complements the fixed-spectrum rotation design. We use the three Gaussian classes and correlation settings in Appendix C.1. Each class is Gaussian; the balanced pooled distribution is a mixture. Joint training uses $n_* \in \{200, 500, 800\}$ observations per class. Under equal target size, both controls use $M = n_*$ target observations; under equal total size, both use $M = 3n_*$ target observations.

Under equal target size, C_{J} has the lowest score risk, validation noise MSE, and Gaussian W_2^2 at every sample size (Table 6). The advantage narrows as target size increases. At $n_* = 200$, for example, joint score risk is 84.90, compared with 1943.75 for U_{T} and 2359.48 for C_{T}; the moment distances are 7.57, 23.81, and 31.32. The additional target-only unconditional control shows that this gain is associated with auxiliary training beyond merely adding a label input.

Under equal total size, joint training again has the lowest aggregate score risk, but both controls have lower Gaussian W_2^2, and U_{T} has the lowest validation noise MSE (Table 7). At $n_* = 200$, the score risks of C_{J} and U_{T} are 84.04 and 99.40, while their Gaussian W_2^2 values are 7.51 and 5.27. These gaps narrow with sample size. Thus a score-risk gain does not determine the ordering of generated-moment metrics.

E.1 Noise-level diagnostics

Figure 4 reevaluates frozen checkpoints under equal total size, $n_* = 200$, and zero class-varying mismatch. It uses 2,048 common held-out observation–noise pairs at each of 1,000 steps. Approximately 99% of the net score-risk benefit is contributed by the first 100 steps. The score-risk benefits relative to C_{T} and U_{T} are 26.42 and 13.97, whereas the raw noise-MSE benefits are 0.00182 and 0.00115. This diagnostic uses a different Monte Carlo stream from the randomly sampled-time AR(1) tables; it localizes gains across noise levels rather than reproducing their aggregate estimates exactly.

E.2 Covariance-level results under equal target size

Table 8 averages metrics within seed over $n_* \in \{200, 500, 800\}$. The three shared-covariance settings and four class-varying levels are reported separately. Target-only controls use common baselines in the class-varying regime, as described in Appendix C.5. Joint training has lower values of all three metrics than both controls in every reported setting. The decrease in joint score risk from zero to strong mismatch should not be interpreted as evidence that mismatch helps: changing the auxiliary correlations also changes their spectra and denoising difficulty.

Table 6: AR(1) results under equal target size. The joint model uses n_* observations per class and each control uses $M = n_*$. Covariance settings are aggregated as in Appendix C.5. Values are means ± standard errors over 20 seeds.

n_*	Model	Score risk	Validation noise MSE	Gaussian W_2^2
200	U_{T}	1943.75 ± 51.89	0.4092 ± 0.0020	23.81 ± 0.45
200	C_{T}	2359.48 ± 62.50	0.4152 ± 0.0018	31.32 ± 1.18
200	C_{J}	84.90 ± 2.55	0.3014 ± 0.0012	7.57 ± 0.05
500	U_{T}	151.53 ± 4.15	0.3092 ± 0.0014	6.66 ± 0.03
500	C_{T}	185.45 ± 4.58	0.3124 ± 0.0012	7.50 ± 0.02
500	C_{J}	18.94 ± 0.66	0.2839 ± 0.0011	2.82 ± 0.04
800	U_{T}	54.90 ± 1.64	0.2914 ± 0.0013	3.80 ± 0.02
800	C_{T}	59.76 ± 1.52	0.2924 ± 0.0011	3.92 ± 0.02
800	C_{J}	13.86 ± 0.49	0.2817 ± 0.0011	2.14 ± 0.03

Table 7: AR(1) results under equal total size. Joint training uses n_* observations from each class; each control uses $M = 3n_*$ target observations. Aggregation and uncertainty are defined as in Table 6.

n_*	Model	Score risk	Validation noise MSE	Gaussian W_2^2
200	U_{T}	99.40 ± 2.74	0.3004 ± 0.0014	5.27 ± 0.03
200	C_{T}	114.18 ± 2.87	0.3023 ± 0.0011	5.74 ± 0.01
200	C_{J}	84.04 ± 2.42	0.3014 ± 0.0012	7.51 ± 0.06
500	U_{T}	22.18 ± 0.75	0.2824 ± 0.0013	2.35 ± 0.02
500	C_{T}	23.34 ± 0.74	0.2832 ± 0.0011	2.31 ± 0.02
500	C_{J}	19.01 ± 0.62	0.2839 ± 0.0011	2.81 ± 0.03
800	U_{T}	15.51 ± 0.55	0.2799 ± 0.0013	1.89 ± 0.02
800	C_{T}	16.09 ± 0.53	0.2811 ± 0.0011	1.89 ± 0.02
800	C_{J}	14.03 ± 0.48	0.2818 ± 0.0011	2.11 ± 0.02

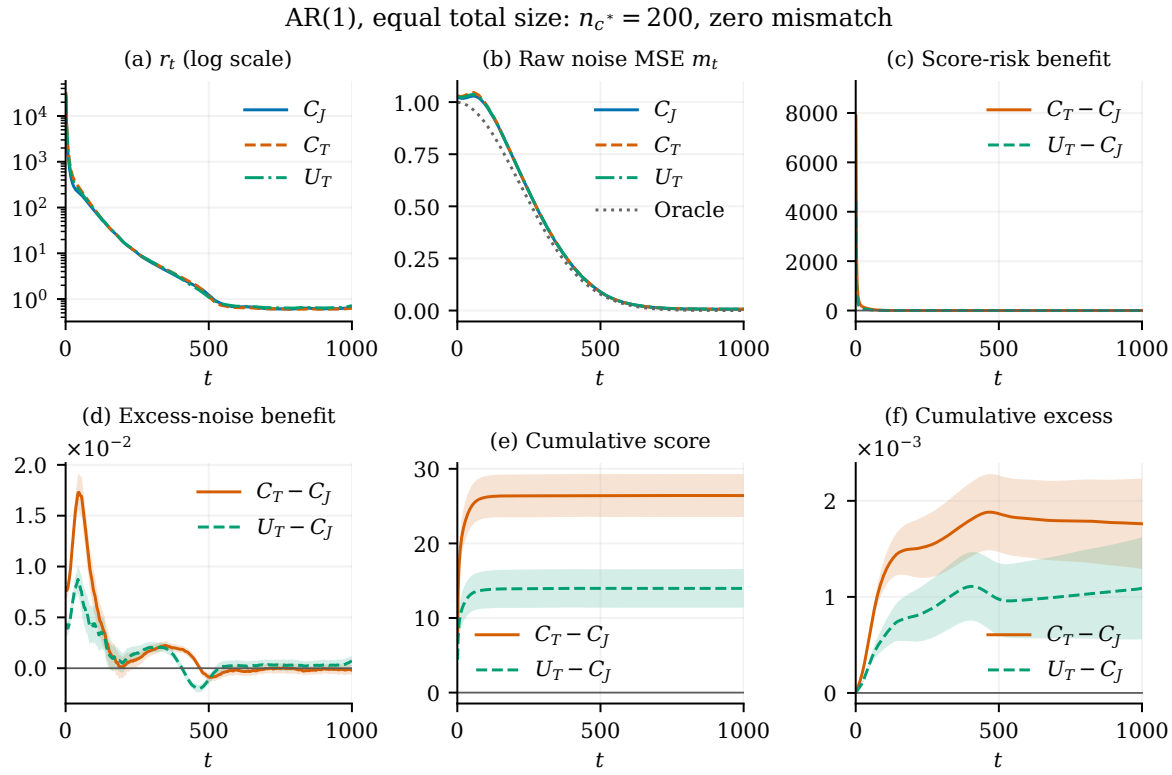


Figure 4: Equal-total AR(1) diagnostics with $n_* = 200$ and zero class-varying mismatch. Panels (a)–(b) show score risk and raw per-coordinate noise MSE. Panels (c)–(f) show control-minus-joint differences, with positive values favoring joint training. Panels (d) and (f) use oracle-excess noise MSE, $d^{-1}\mathbb{E}\,\|\widehat{\varepsilon} - \mathbb{E}[\varepsilon \mid X_t]\|^2$. Cumulative sums are divided by 1,000. Bands are pointwise 95% Student-t intervals across 20 training seeds.

Table 8: AR(1) equal-target results by covariance setting, averaged equally over the three sample sizes within each seed. Values are means ± standard errors over 20 seeds.

Covariance setting	Model	Score risk	Validation noise MSE	Gaussian W_2^2
Shared, $\rho = 0.2$	U_{T}	708.79 ± 18.26	0.3349 ± 0.0016	11.18 ± 0.14
Shared, $\rho = 0.2$	C_{T}	860.82 ± 20.99	0.3383 ± 0.0013	14.41 ± 0.94
Shared, $\rho = 0.2$	C_{J}	40.79 ± 1.18	0.2882 ± 0.0012	4.47 ± 0.03
Shared, $\rho = 0$	U_{T}	730.48 ± 20.39	0.3388 ± 0.0016	11.66 ± 0.36
Shared, $\rho = 0$	C_{T}	881.86 ± 24.11	0.3419 ± 0.0014	15.30 ± 1.07
Shared, $\rho = 0$	C_{J}	41.15 ± 1.52	0.2902 ± 0.0012	3.87 ± 0.05
Shared, $\rho = -0.2$	U_{T}	706.19 ± 19.27	0.3350 ± 0.0016	11.21 ± 0.12
Shared, $\rho = -0.2$	C_{T}	852.39 ± 23.51	0.3384 ± 0.0013	13.29 ± 0.27
Shared, $\rho = -0.2$	C_{J}	40.73 ± 1.31	0.2881 ± 0.0011	4.44 ± 0.03
Zero mismatch	C_{J}	40.74 ± 1.22	0.2900 ± 0.0011	4.07 ± 0.04
Mild	C_{J}	38.15 ± 1.24	0.2899 ± 0.0011	4.01 ± 0.04
Medium	C_{J}	32.68 ± 1.13	0.2894 ± 0.0011	3.88 ± 0.05
Strong	C_{J}	25.48 ± 0.83	0.2891 ± 0.0011	3.73 ± 0.04
Common target-only	C_{T}	877.85 ± 23.39	0.3415 ± 0.0014	13.99 ± 0.44
Common target-only	U_{T}	721.43 ± 19.04	0.3377 ± 0.0015	11.65 ± 0.21

Table 9: AR(1) equal-total results by covariance setting, averaged equally over the three sample sizes within each seed. Values are means ± standard errors over 20 seeds. Common target-only baselines combine the four class-varying evaluations.

Covariance setting	Model	Score risk	Validation noise MSE	Gaussian W_2^2
Shared, $\rho = 0.2$	U_{T}	45.83 ± 1.42	0.2866 ± 0.0013	3.41 ± 0.03
Shared, $\rho = 0.2$	C_{T}	51.55 ± 1.41	0.2881 ± 0.0011	3.55 ± 0.02
Shared, $\rho = 0.2$	C_{J}	41.01 ± 1.32	0.2883 ± 0.0011	4.42 ± 0.03
Shared, $\rho = 0$	U_{T}	45.31 ± 1.27	0.2886 ± 0.0014	2.81 ± 0.02
Shared, $\rho = 0$	C_{T}	50.50 ± 1.38	0.2899 ± 0.0011	2.97 ± 0.01
Shared, $\rho = 0$	C_{J}	40.40 ± 1.03	0.2903 ± 0.0012	3.87 ± 0.04
Shared, $\rho = -0.2$	U_{T}	45.56 ± 1.38	0.2867 ± 0.0014	3.43 ± 0.02
Shared, $\rho = -0.2$	C_{T}	52.11 ± 1.46	0.2882 ± 0.0011	3.54 ± 0.02
Shared, $\rho = -0.2$	C_{J}	40.51 ± 1.34	0.2881 ± 0.0011	4.38 ± 0.05
Zero mismatch	C_{J}	40.50 ± 1.20	0.2898 ± 0.0011	4.06 ± 0.05
Mild	C_{J}	38.34 ± 1.42	0.2897 ± 0.0011	4.00 ± 0.06
Medium	C_{J}	32.45 ± 1.14	0.2893 ± 0.0011	3.87 ± 0.06
Strong	C_{J}	25.37 ± 0.78	0.2888 ± 0.0011	3.67 ± 0.04
Common target-only	C_{T}	50.66 ± 1.33	0.2894 ± 0.0011	3.19 ± 0.01
Common target-only	U_{T}	46.07 ± 1.33	0.2885 ± 0.0013	3.03 ± 0.02

E.3 Covariance-level results under equal total size

Table 9 uses the same covariance-level aggregation under equal total size. Joint training has the lowest score risk in every setting, whereas the unconditional target-only control has the lowest validation noise MSE and Gaussian W_2^2. The target distribution remains fixed within the class-varying regime; only auxiliary covariances change.

E.4 Representative PCA projections

Figure 5 compares generated target samples from joint and unconditional target-only training with true target samples. These two-dimensional projections are descriptive views of 100-dimensional distributions; the quantitative conclusions use score risk and Gaussian moment distance.

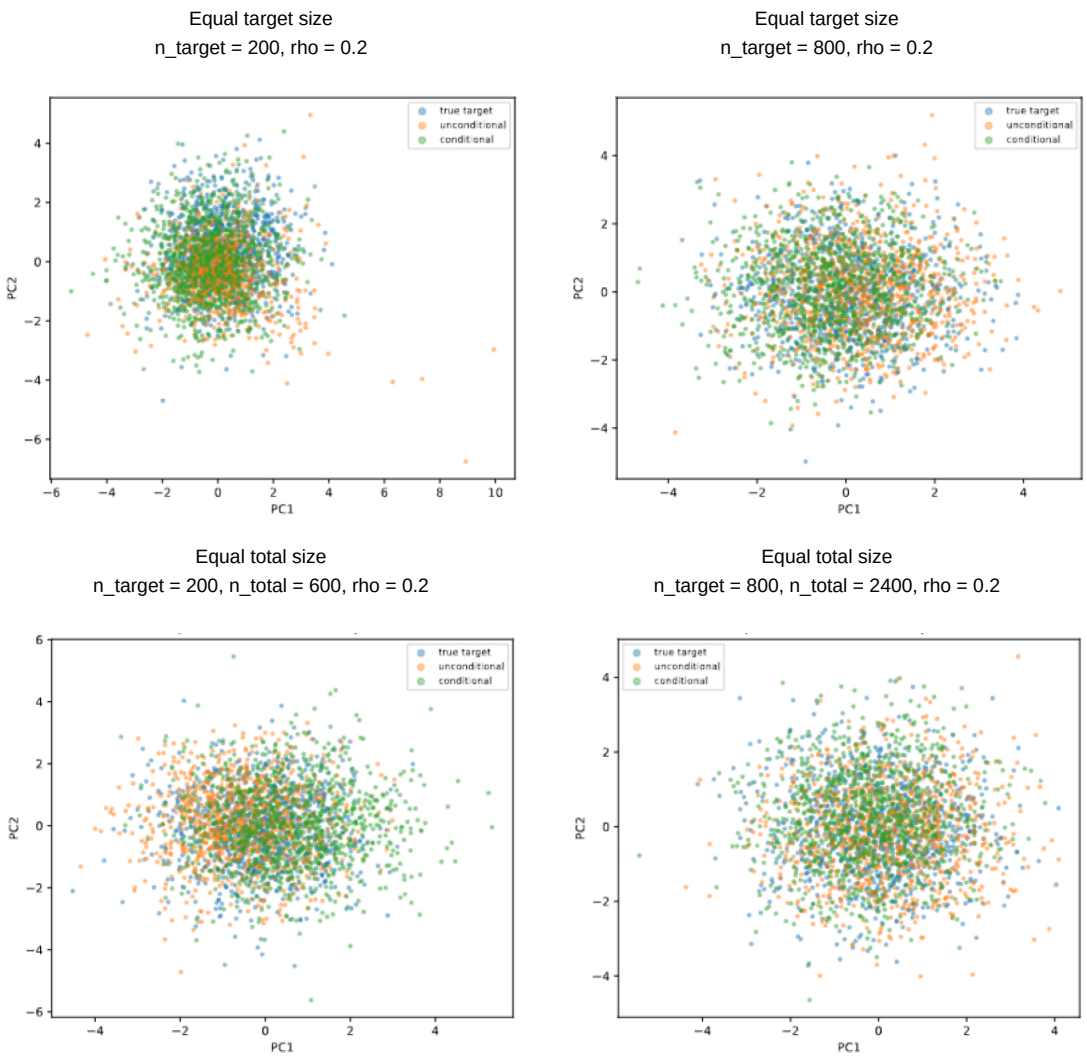


Figure 5: Representative PCA projections under equal target size (top) and equal total size (bottom), with $n_* = 200$ and 800. Bottom-row total sizes are $n = M = 600$ and 2400. All panels use seed 0 and shared correlation $\rho = 0.2$; principal components are fitted to pooled true and generated samples within each panel. The scatter points are retained from the original experiment.

F Real-image experimental details

Data and allocation. The target is French bulldog (n02108915). Close auxiliaries are Boston bull (n02096585), pug (n02110958), and boxer (n02108089); far auxiliaries are school bus (n04146614), church (n03028079), and teapot (n04398044). Images are converted to RGB, resized bilinearly to shorter-side length 64, center-cropped to 64×64, and scaled to $[-1, 1]$.

Under equal target size, the 250-image training set is a fixed subset of the 500-image set, using subset seed 250. All six models share 100 target validation images and 600 target test images. Under equal total size, joint class counts, with the target first, are $(850, 50, 50, 50)$, $(750, 84, 83, 83)$, and $(500, 167, 167, 166)$. The control uses 1,000 target images; joint target subsets come from the same pool. This design has separate 100-image validation and test sets. Training, validation, and test sets are disjoint within each design, and comparisons use common evaluation images.

Architecture and training. We use an EDM-preconditioned SongUNet with approximately 15.7 million parameters, base width 64, channel multipliers $(1, 2, 2, 2)$, four residual blocks per resolution, and attention at 16×16. It uses Fourier noise embeddings, a residual encoder, a standard decoder, and dropout 0.13. The target-only model has one class label and each joint model has four. Shared weights and effective target-label conditioning are initialized identically. All models are trained from scratch with seed 0.

Adam uses learning rate 2×10^{-4}, $(\beta_1, \beta_2) = (0.9, 0.999)$, numerical constant 10^{-8}, and 1,000 warmup updates. Batch size is 64, accumulated in microbatches of 16, with FP32 network computation. The EMA half-life is 50,000 processed images, with rampup ratio 0.05. All models use the same EDM geometric augmentations, with probability parameter 0.12 and nine augmentation-conditioning coordinates. Images are sampled uniformly with replacement from the merged training pool. Thus equal-target joint training draws target images with probability 1/4; the primary comparison matches update count and batch size, rather than the number of target-image presentations.

Denoising evaluation. We use additive Gaussian corruption $X_t^* = X_0^* + t\varepsilon$, with $d = 3 \cdot 64^2$, $\log t \sim \mathcal{N}(-1.2, 1.2^2)$, and $\sigma_{\text{data}} = 0.5$. The reported per-pixel criterion is

$$\frac{1}{d}\mathbb{E}_{t,X_0^*,\varepsilon}\left[\lambda_D(t)\,\|D_{\theta,u}(X_t^*, t) - X_0^*\|^2\right], \qquad \lambda_D(t) = \frac{t^2 + \sigma_{\text{data}}^2}{t^2\sigma_{\text{data}}^2}. \tag{F.1}$$

Evaluation uses no augmentation. Equal-target evaluation averages eight noise draws per image; equal-total evaluation uses two. Images, noise levels, and Gaussian-noise realizations are paired across models within each design. Training uses the same noise law and weight, summing over pixels and averaging over examples; the reported metric additionally divides by d.

By (B.7), population differences in (F.1) equal $1/d$ times target score-risk differences under

$$w(t) = \nu(t)\lambda_D(t)t^4 = \nu(t)\left(t^2 + \frac{t^4}{\sigma_{\text{data}}^2}\right), \qquad t > 0,$$

where ν is the lognormal density. This identity explains the score-risk interpretation of the reported weighted denoising metric; it does not equate that metric with unweighted score risk.

Sampling and generated-image metrics. EMA models generate 2,048 target images with the EDM Heun sampler, 40 noise levels, minimum 0.002, maximum 80, schedule exponent 7, and zero stochastic churn. Initial-noise seed 123456 is shared by all models. Sampling uses neither classifier-free guidance nor denoised-prediction clipping. The ODE state uses FP64 and the

Table 10: Test denoising MSE after checkpoint selection by validation MSE. Training size includes all classes. Equal-target evaluation uses 600 common test images and eight paired draws per image; equal-total evaluation uses 100 common test images and two draws per image. Every model uses training seed 0.

Training size	Model	Target fraction	Updates	Test MSE
Equal target size: $n_* = 250$				
250	C_{T}	1.00	10000	0.148815
1000	$C_{\mathrm{J}}^{\mathrm{close}}$	0.25	30000	0.139769
1000	$C_{\mathrm{J}}^{\mathrm{far}}$	0.25	30000	0.140663
Equal target size: $n_* = 500$				
500	C_{T}	1.00	10000	0.144090
2000	$C_{\mathrm{J}}^{\mathrm{close}}$	0.25	50000	0.136975
2000	$C_{\mathrm{J}}^{\mathrm{far}}$	0.25	70000	0.137505
Equal total size: $n = M = 1000$				
1000	C_{T}	1.00	60000	0.140994
1000	$C_{\mathrm{J}}^{\mathrm{close}}$	0.85	60000	0.140585
1000	$C_{\mathrm{J}}^{\mathrm{close}}$	0.75	60000	0.140646
1000	$C_{\mathrm{J}}^{\mathrm{close}}$	0.50	60000	0.141595

denoiser uses FP32. Outputs are quantized by $\mathrm{clip}(127.5x + 128, 0, 255)$ followed by unsigned 8-bit conversion.

KID and FID use 2,048-dimensional Inception-v3 features from the TensorFlow-compatible `torch-fidelity` extractor. KID is the full-sample unbiased squared MMD with kernel $k(f, g) = (f^\top g/2048 + 1)^3$. Image tables report 10^3KID and unscaled FID. FID uses empirical feature means and covariances, computed through low-rank factors. Both metrics use all test and generated images, with a common reference set within each design. Reference-set sizes differ between the two designs, so their absolute KID and FID values should be compared within a design.

Validation-selected checkpoints. The primary results use the fixed 80,000-update endpoint. A secondary comparison selects checkpoints by target validation MSE, breaking ties in favor of the earlier checkpoint. Common candidates are 10,000, 20,000, ..., 80,000 updates under equal target size and $\{60000, 80000\}$ under equal total size. Selection precedes test evaluation. Table 10 reports the resulting test MSE. Both joint models retain lower MSE than the target-only control under equal target size. All comparisons still use one training seed.



Figure 6: First 64 saved target-class images from $C_{\mathrm{J}}^{\mathrm{close}}$ under equal target size, $n_* = 500$. Images are ordered left to right and top to bottom, from the 80,000-update EMA checkpoint with the 40-level EDM Heun sampler and initial-noise seed 123456. Each image is 64×64.